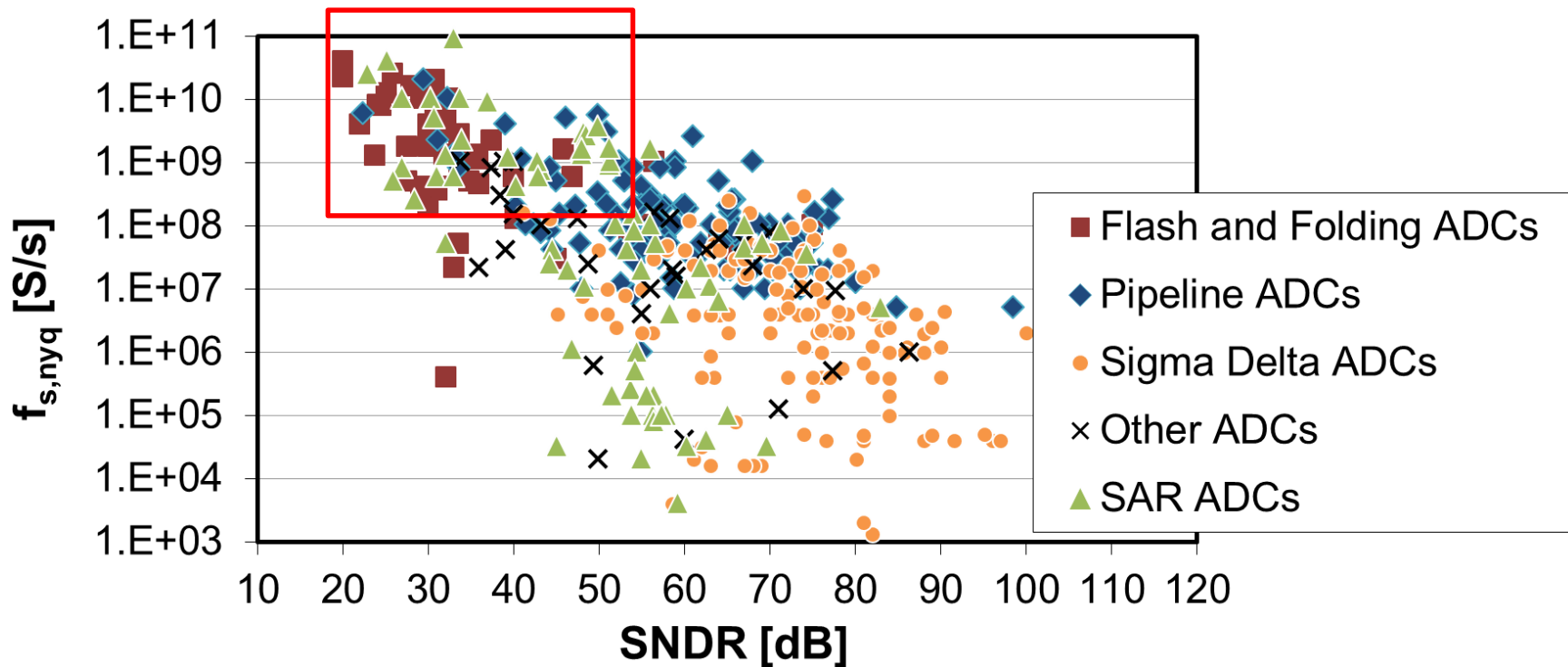


High-Speed SAR ADC Design

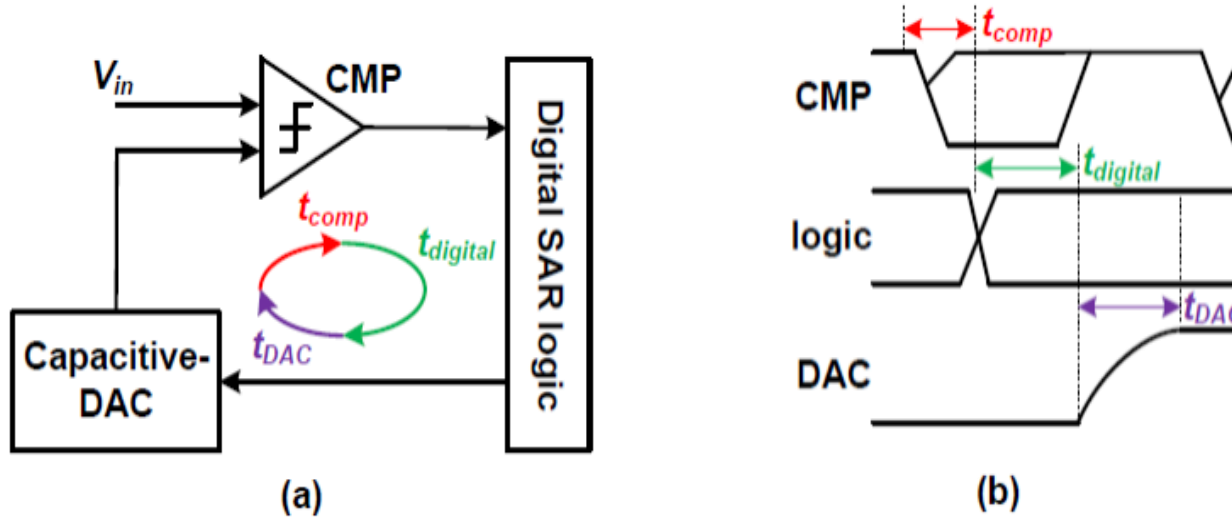
Nan Sun
Tsinghua University
nansun@tsinghua.edu.cn

High speed SAR ADCs



- Common to see SAR ADCs with sampling rates $> 100\text{MS/s}$
 - Single channel $> 1\text{GS/s}$
 - Time interleaved SAR ADC $\sim 100\text{GS/s}$
- What makes it possible?
 - Technology scaling
 - Circuit ingenuity

Classic SAR



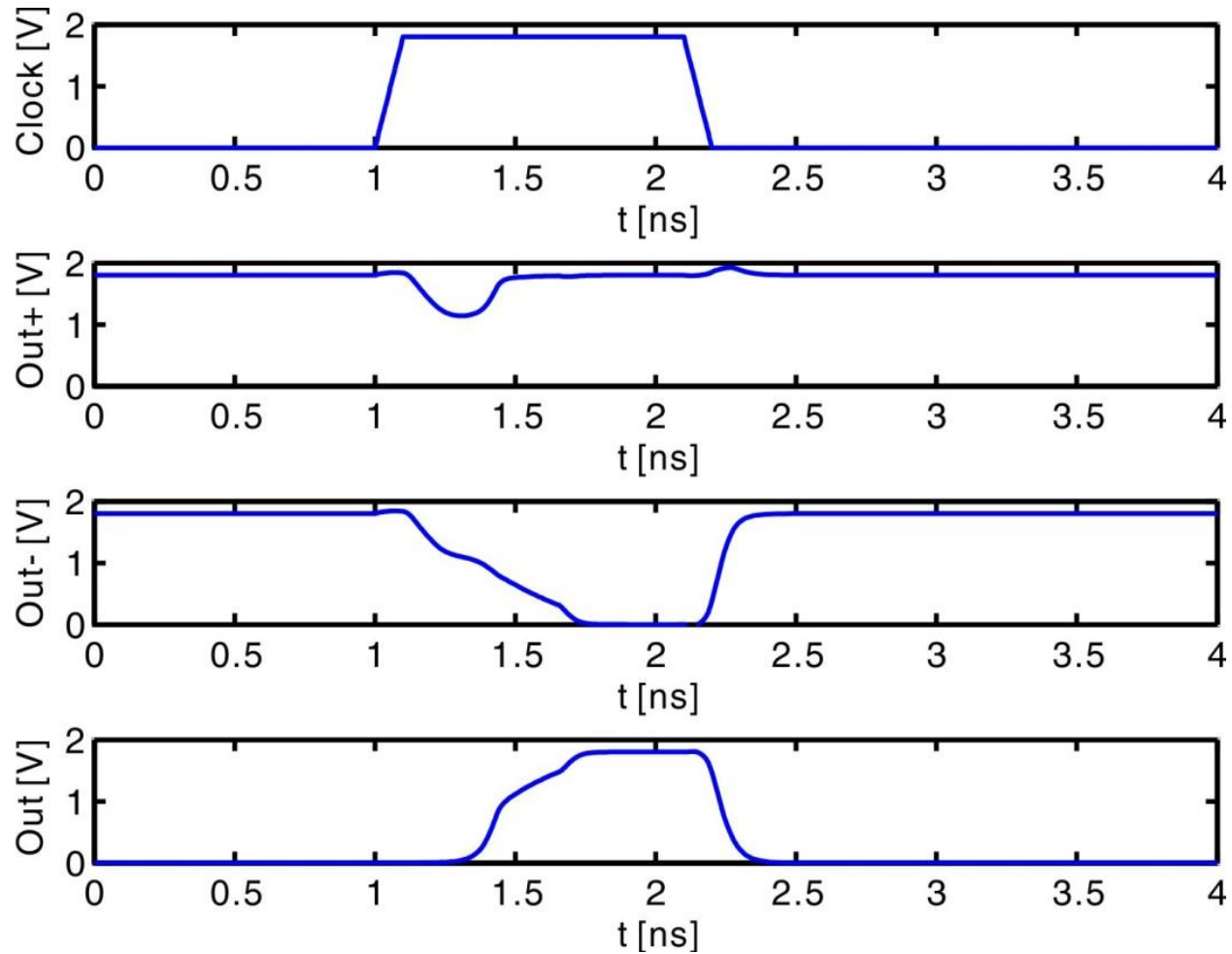
$$T_{loop} = t_{comp} + t_{logic} + \max\{t_{DAC}, t_{reset}\}$$

$$T_{SAR} \approx t_{sample} + N \cdot T_{loop}$$

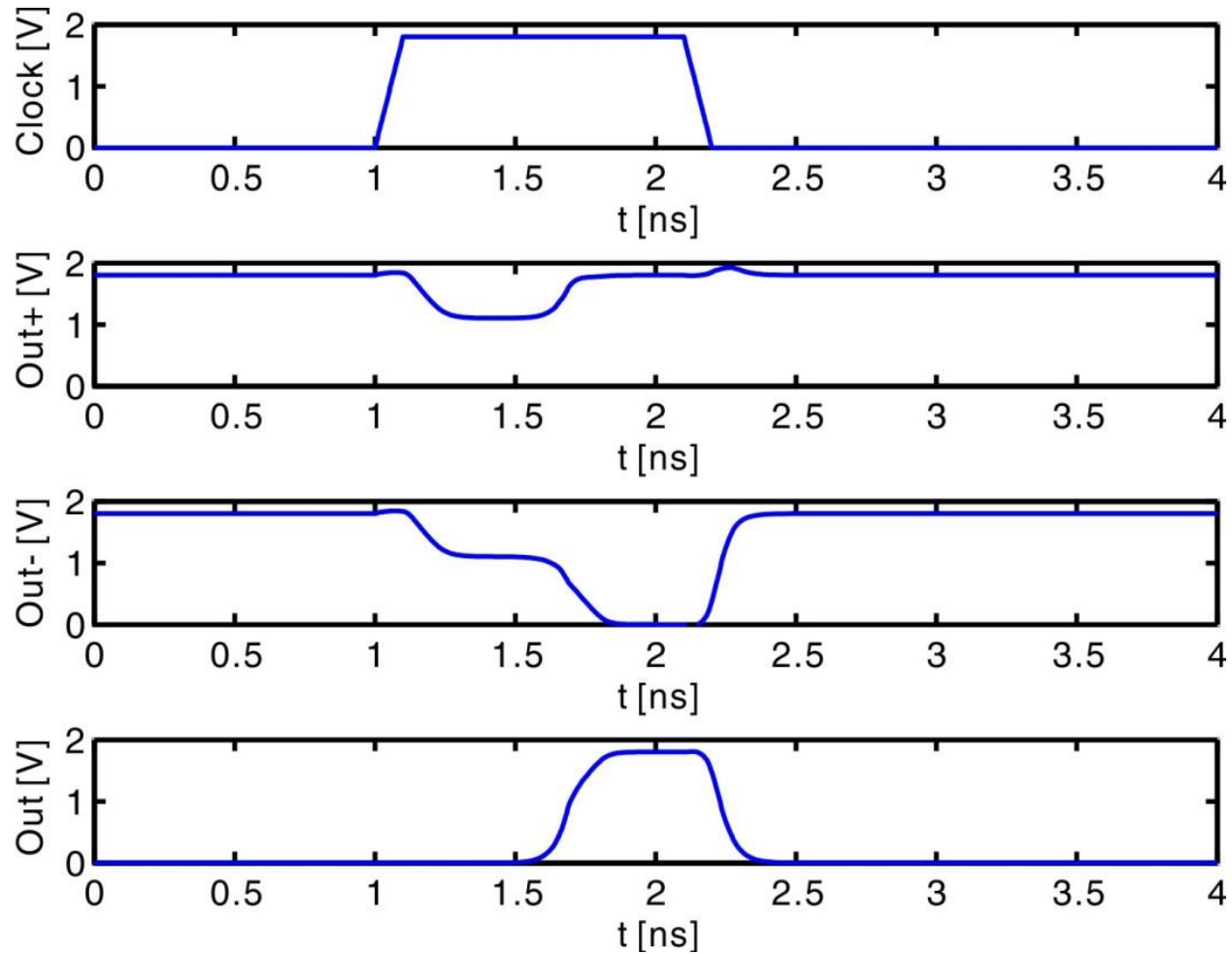
High-speed SAR design

- $T_{loop} = t_{comp} + t_{logic} + \max\{t_{DAC}, t_{reset}\}$
- $T_{SAR} \approx t_{sample} + N \cdot T_{loop}$
- Reducing t_{comp}
 - Asynchronous clocking
- Reducing t_{logic}
 - Direct path from comparator to DAC
- Reducing t_{DAC}
 - Add redundancy
 - Self timing
- Reducing t_{reset} ($t_{reset} > t_{DAC}$ for $b < 6$ -bit)
 - Ping-pong operation
 - Loop unrolled SAR
- Reducing N
 - Multi-bit-per-cycle

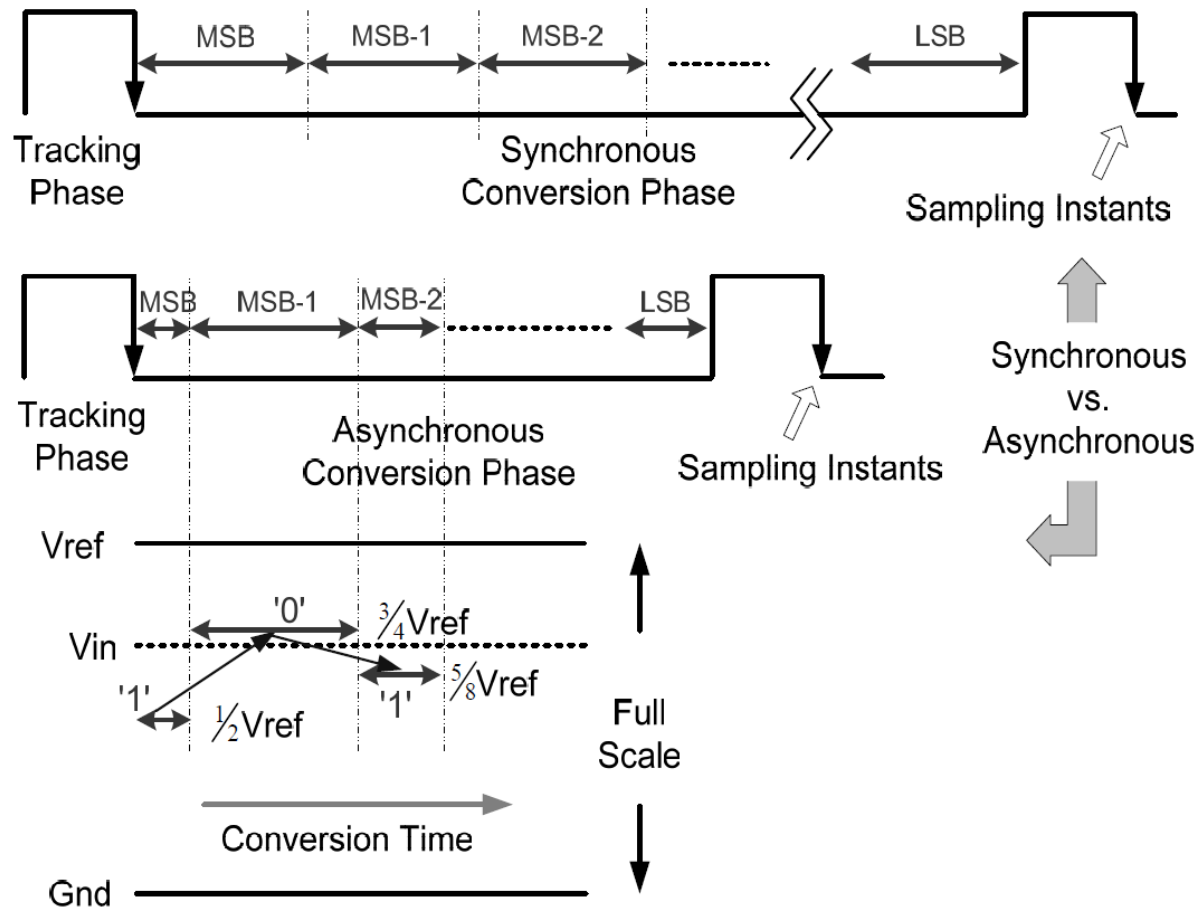
Simulation for $V_{in} = 1\text{mV}$



Simulation for $V_{in} = 1\mu V$

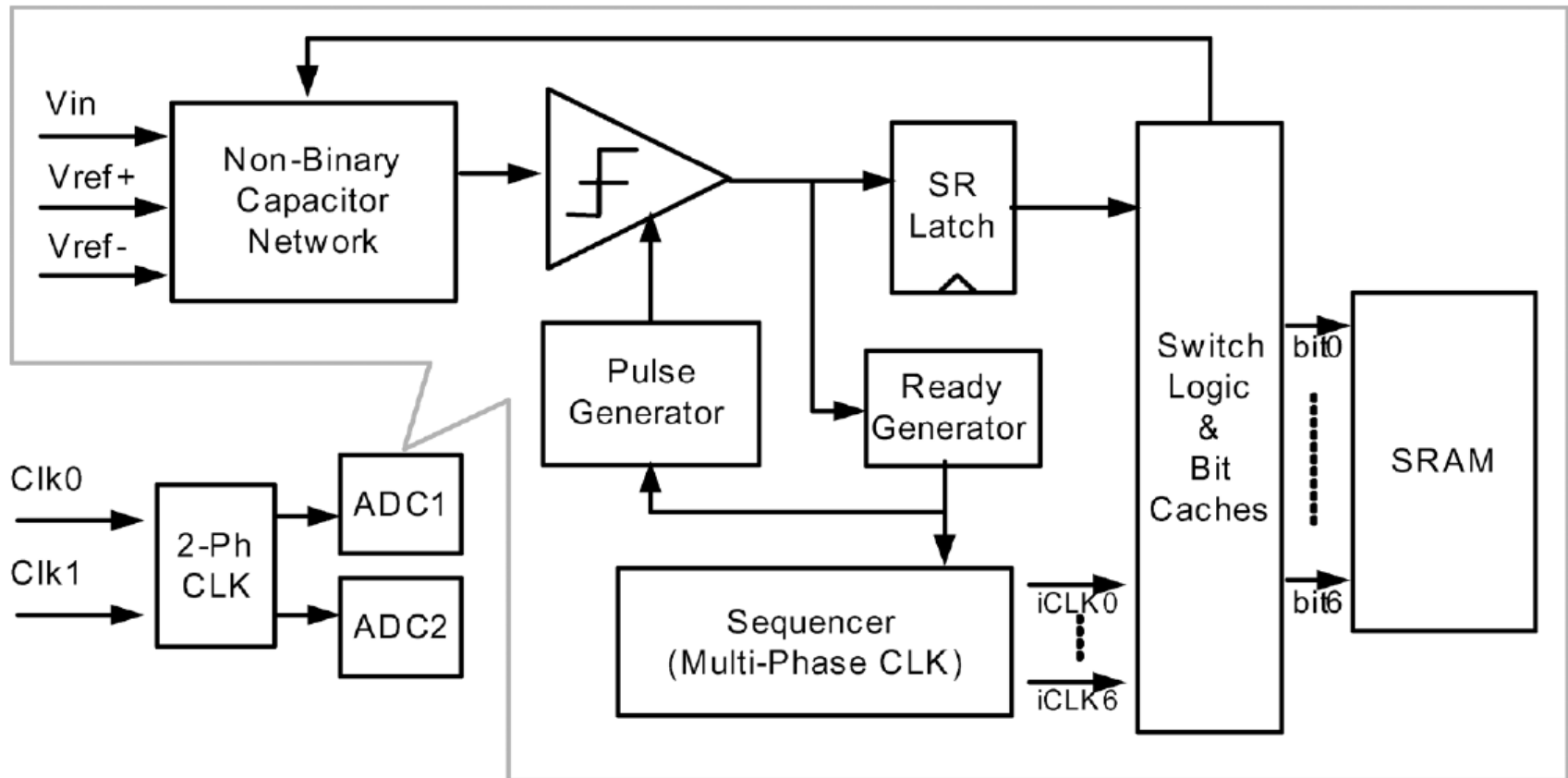


Reducing total comparator delay



- Comparator delay is signal dependent $\rightarrow$ smaller input takes longer time to resolve
- Only one comparison has input less than $\Delta/2$; do not budget for the worst case

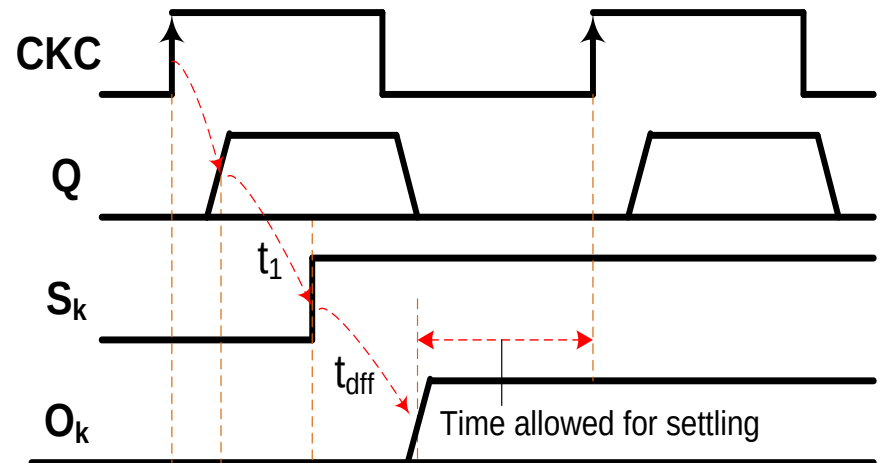
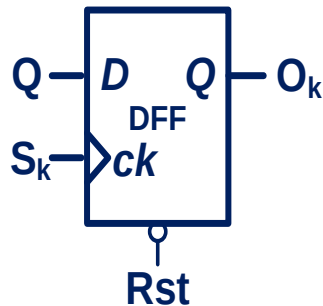
Asynchronous SAR Architecture



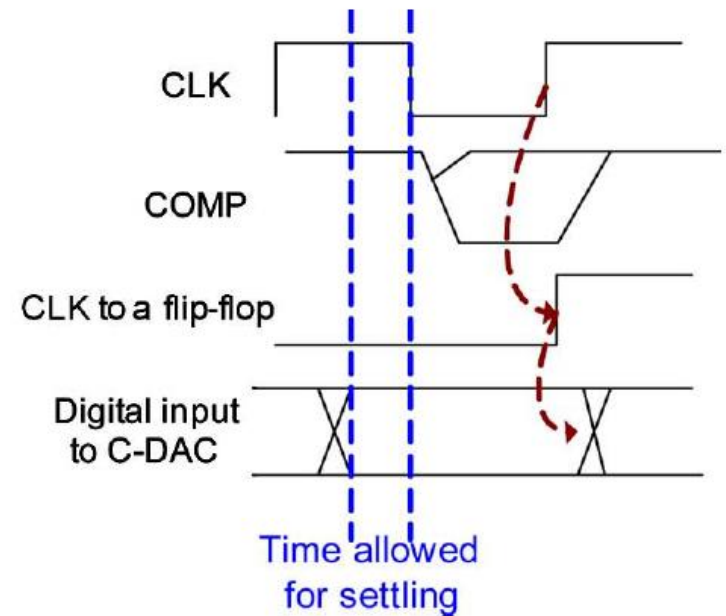
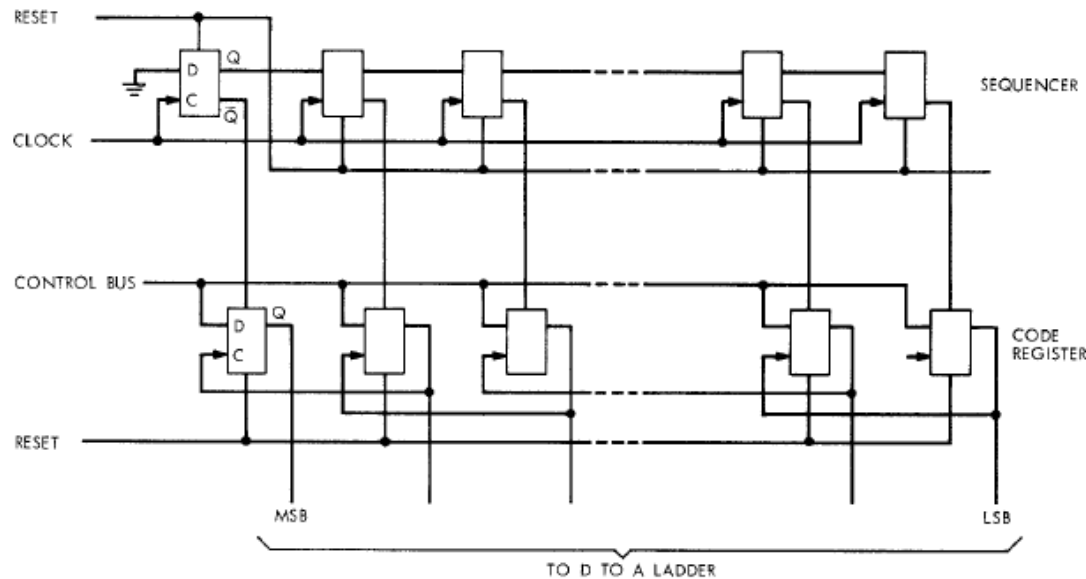
A NAND gate is sufficient as a comparator ready generator

Reducing logic delay

- Conventional SAR logic:
 - Comparator ready $\rightarrow$ sequencer $\rightarrow$ data-storage $\rightarrow$ DAC
 - Long delay from comparator output to DAC update

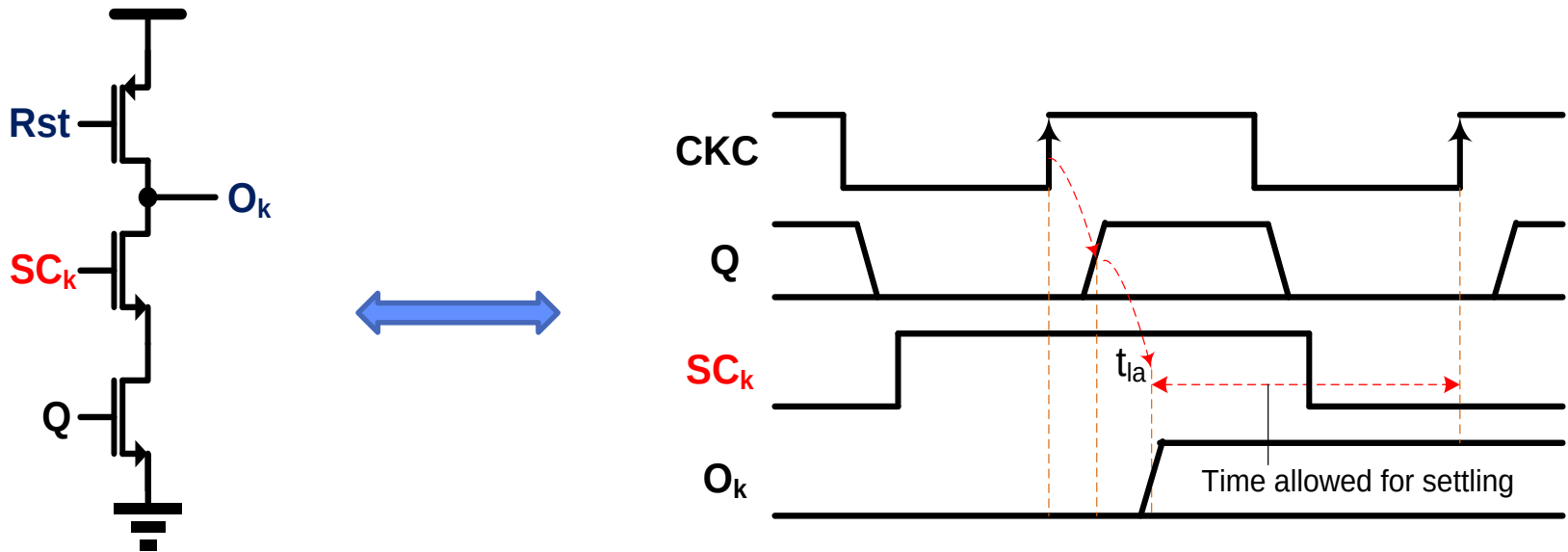


Typical SAR Logic and Timing Diagram

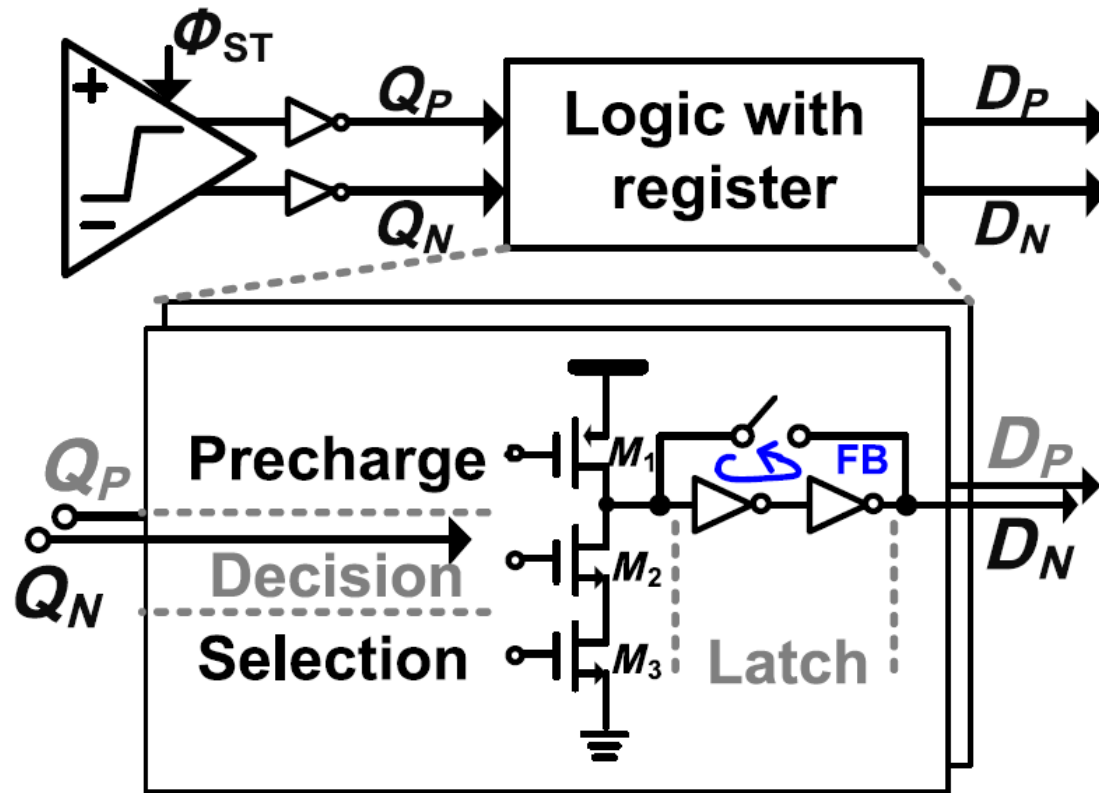


Fast dynamic SAR logic

- Sequencer state (SC_k) pre-determined
- Comparator output (Q) directly goes to DAC (O_k) via transparent latch
- Very fast logic, but can have metastability issue (more later)

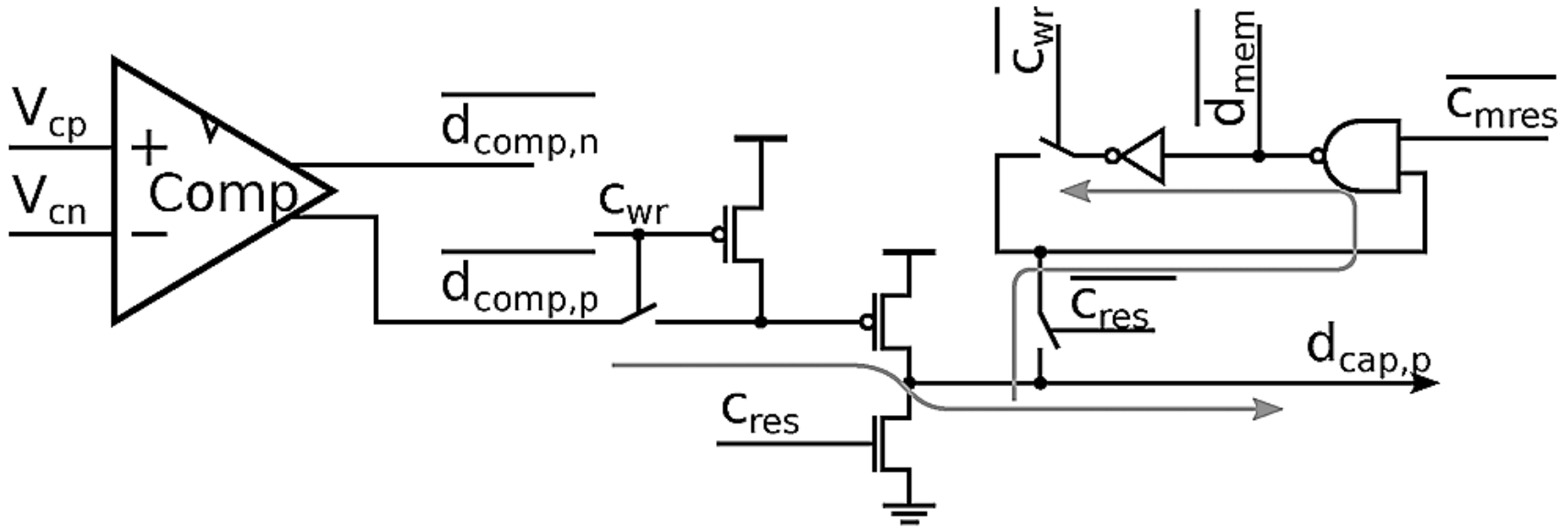


High-speed SAR logic example



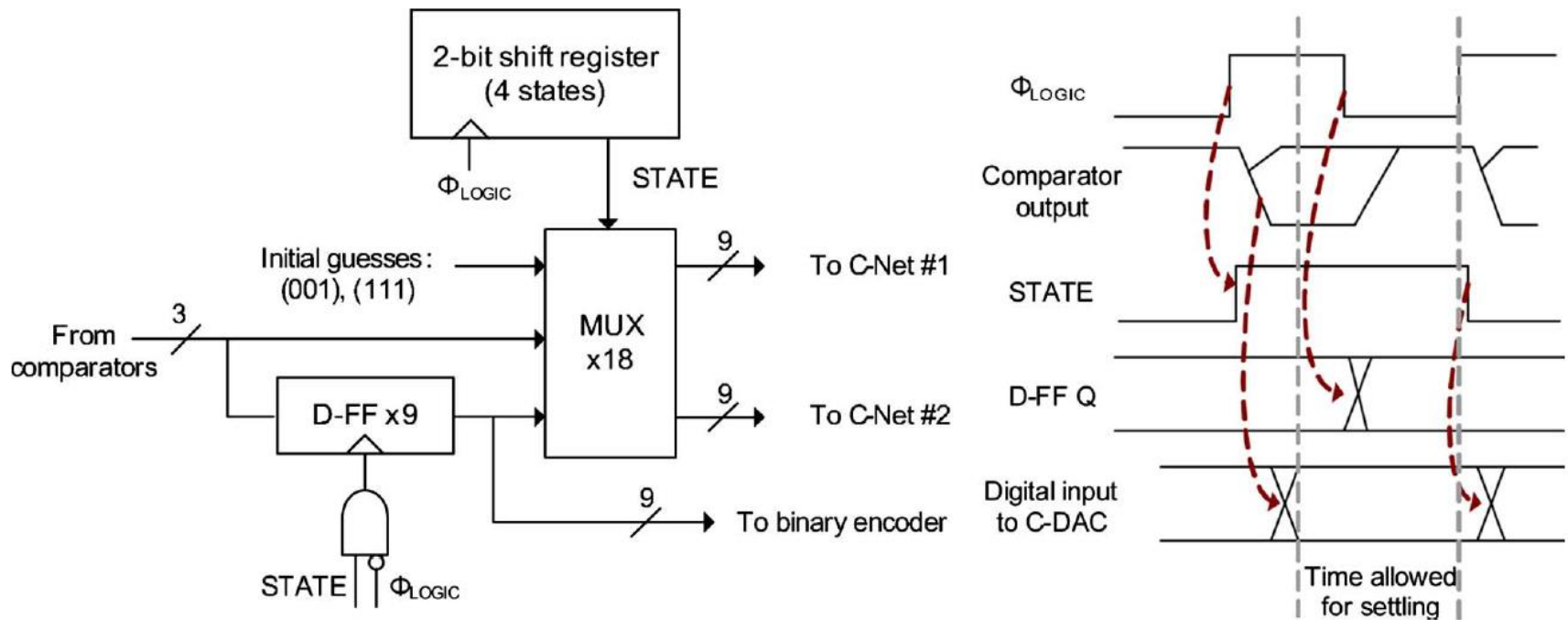
- Sequencer state pre-determined
- Comparator output directly goes to the DAC
- Data is latched during DAC settling

High-speed SAR logic example



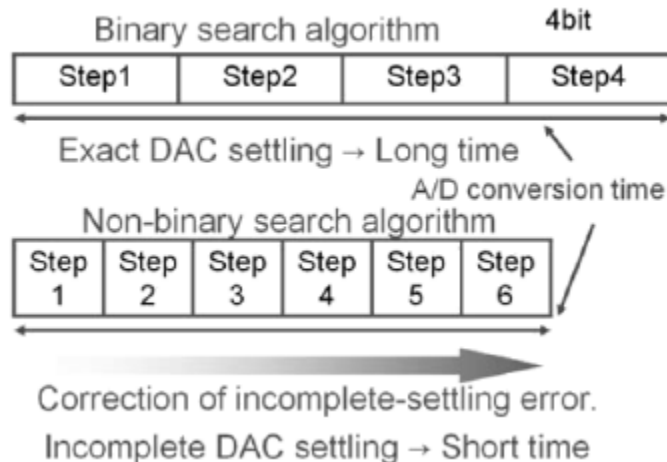
- Sequencer state pre-determined
- Comparator output directly goes to the DAC
- Data is latched during DAC settling

Fast SAR logic



Reducing t_{DAC}

- Without redundancy, each bit needs to settle to the target resolution
 - For a 10-bit SAR, need to allocate 7τ
- With redundancy, each bit only needs to settle to the redundancy range
 - Incomplete settling error can be recovered later
 - For a 10-bit SAR with 5% redundancy, only need 3τ

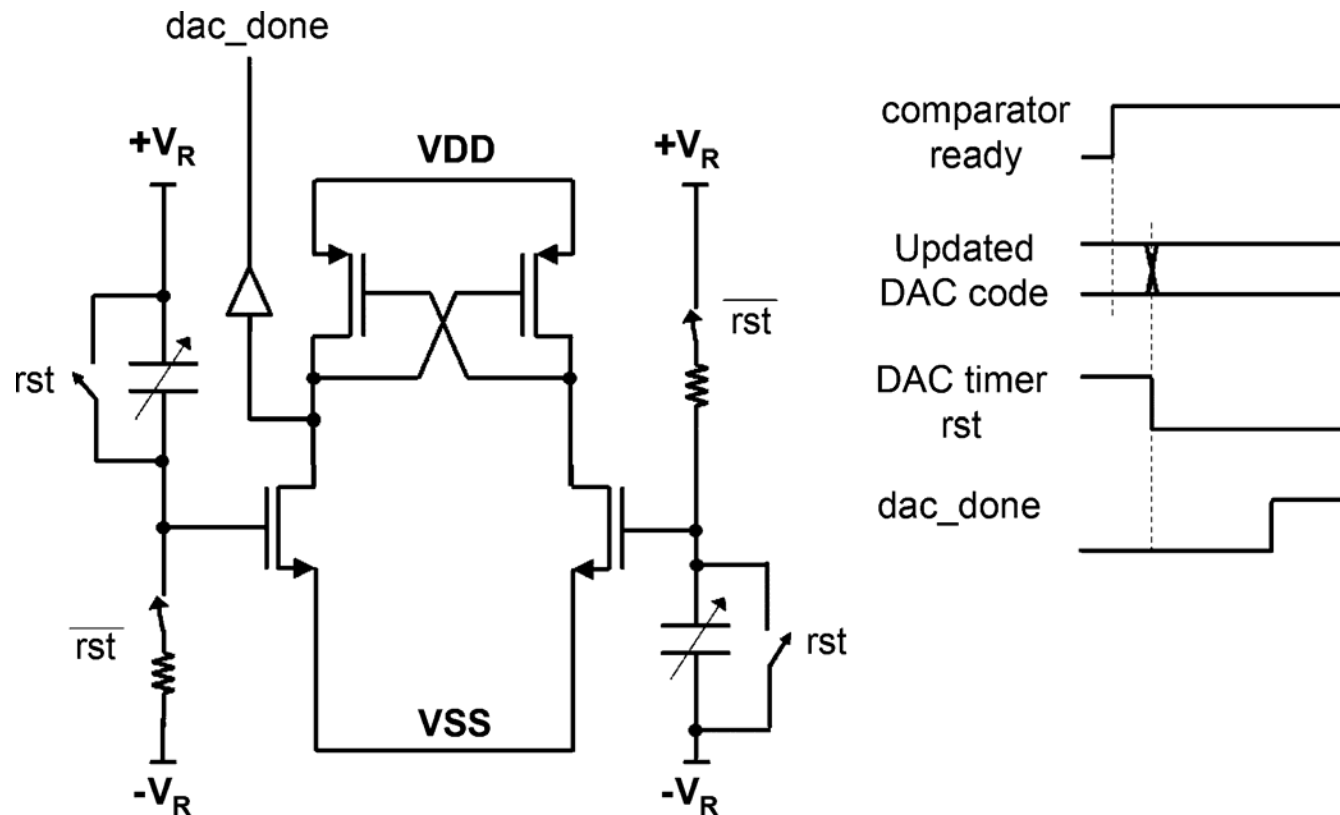


Algorithm	Binary	Prior	Generalized
Time slot for each step	6.3τ	2.2τ	1.9τ
Number of steps	10	12	12
Total conversion time	56.7τ	24.2τ	20.9τ

T. Ogawa et al., IEICE Trans. on Fundamentals of Electronics, Communications and Computer Sciences, Feb. 2010.

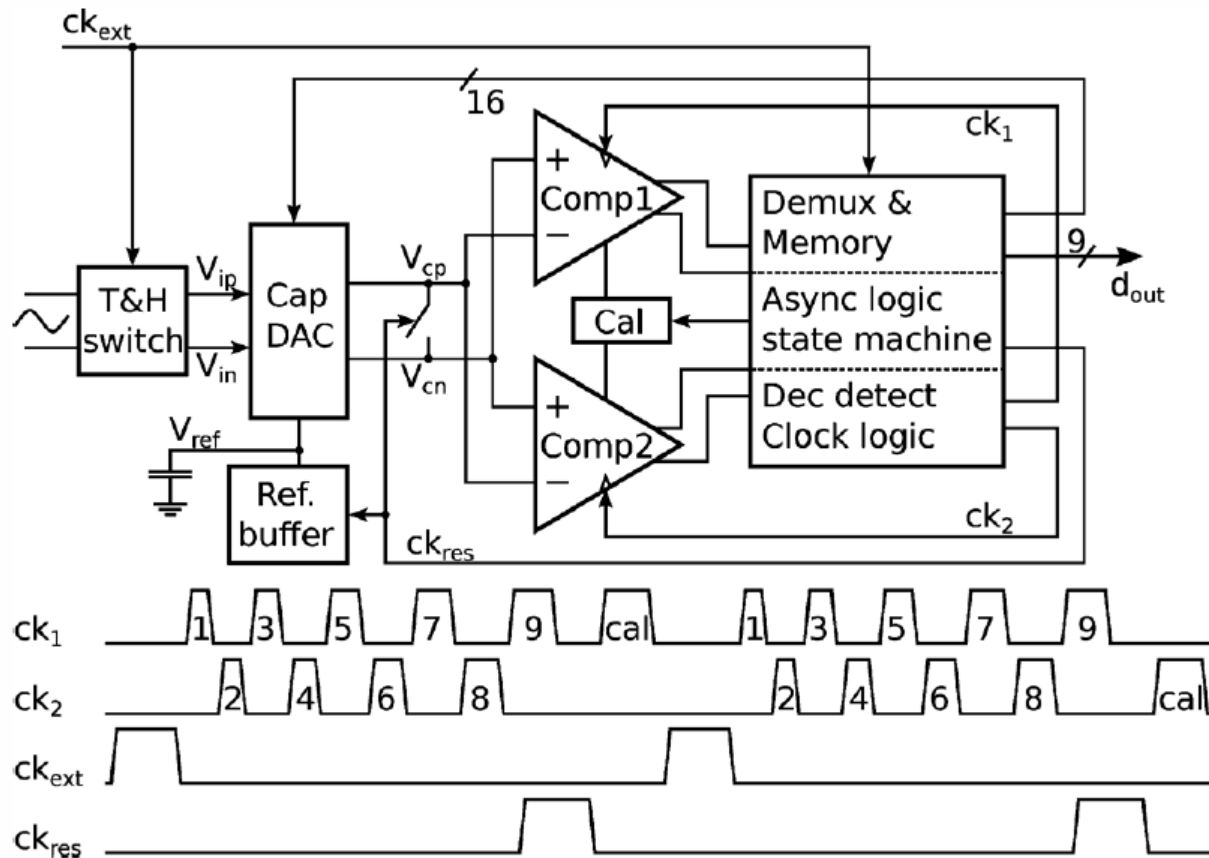
DAC self-timing

- DAC settling speed also depends on PVT
- Also depend on redundancy and bits
- Having a replica prevents over-design

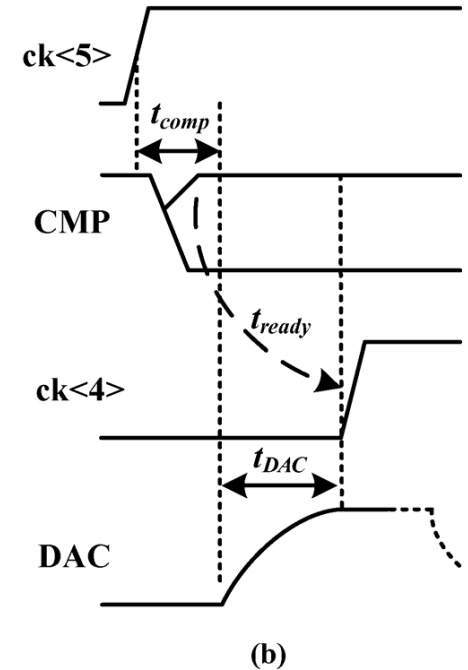
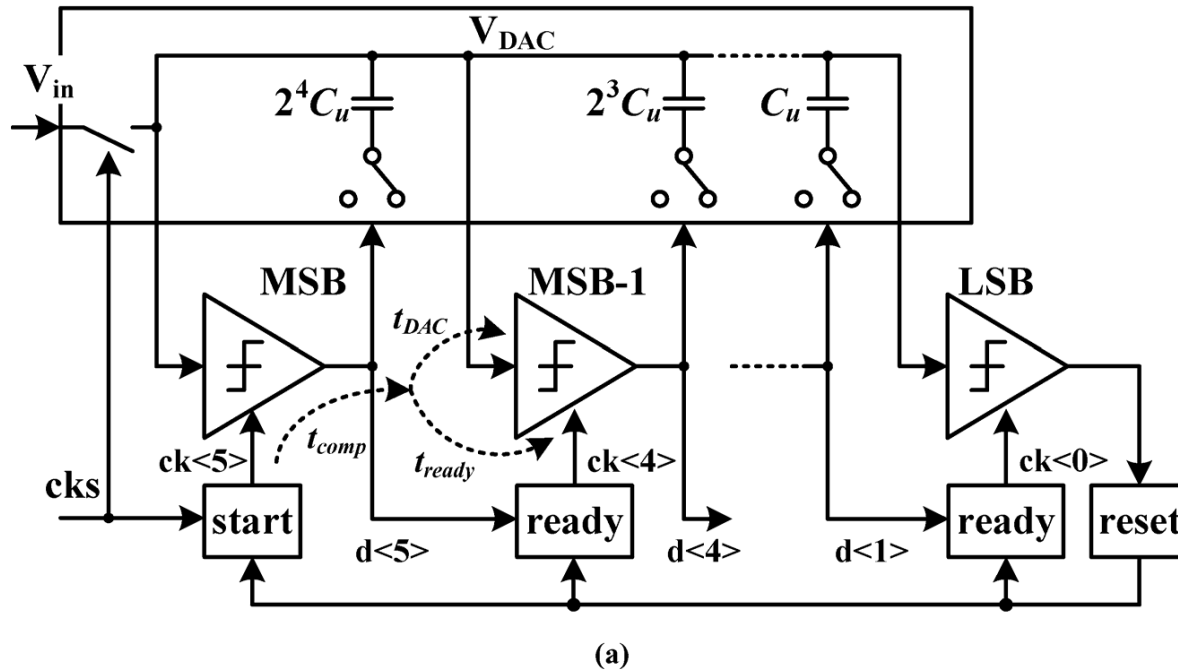


Reducing t_{reset} : ping-pong comparator

- Time interleave 2 comparators
 - Fire one comparator and reset the other at the same time
 - Only need to calibrate 2 comparator offset

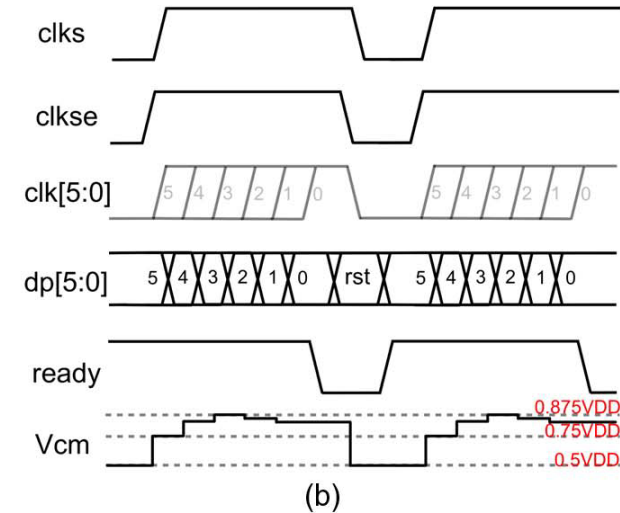
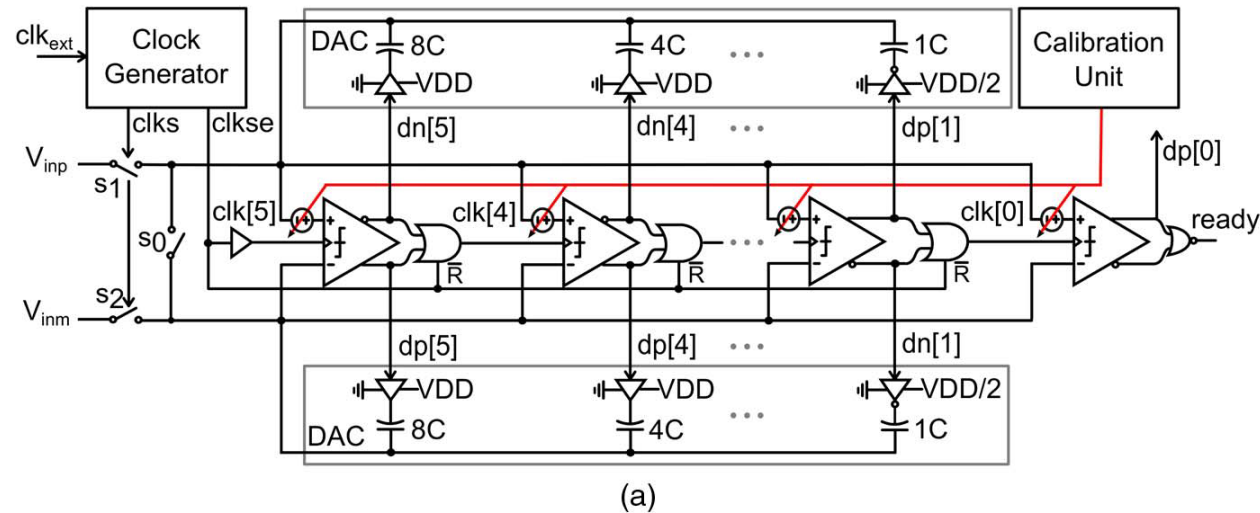


Loop-Unrolled SAR



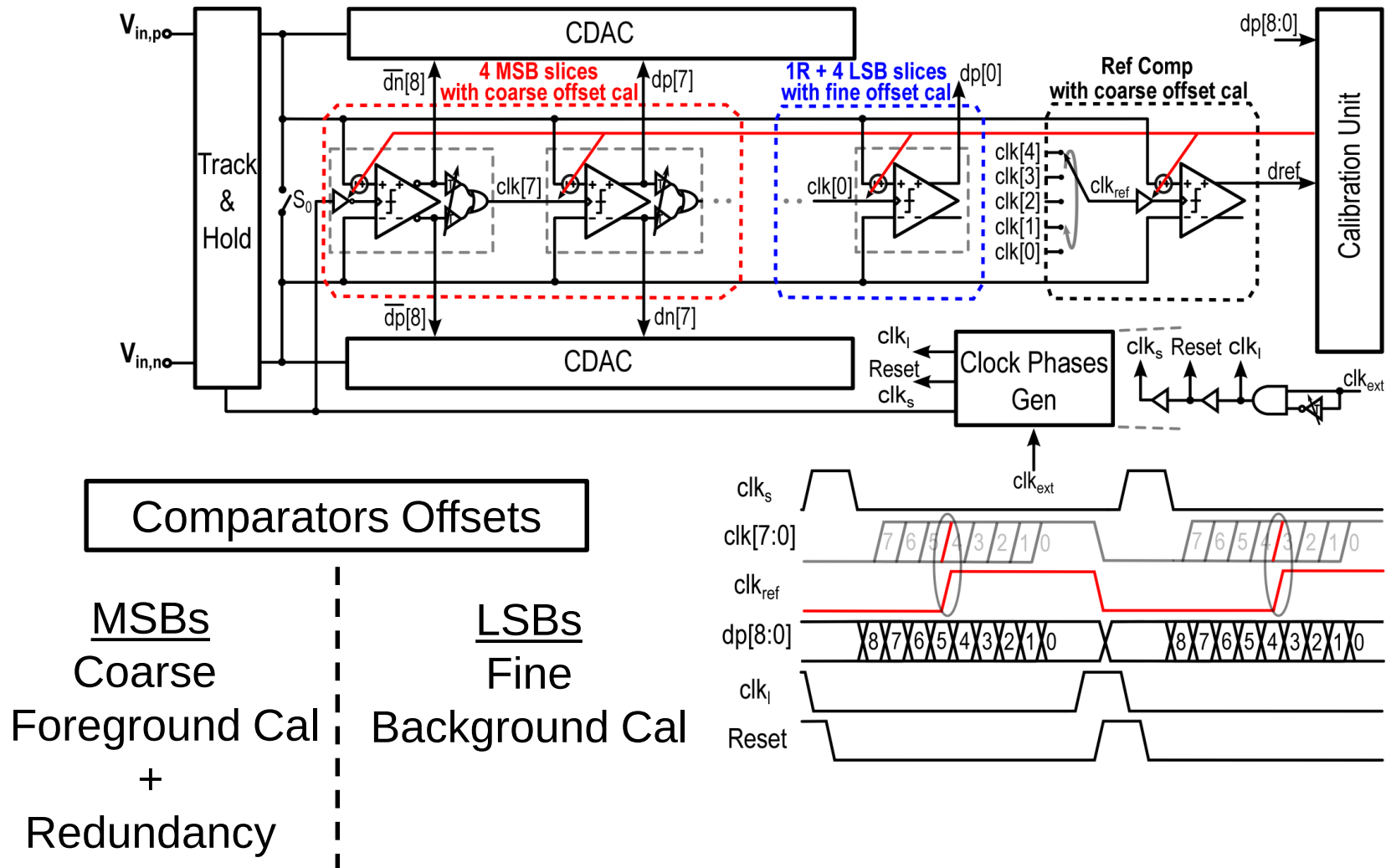
- $T_{critical} = t_{comp} + t_{DAC}$ 😊
- No logic delay 😊; No comparator reset 😊
- Comparator offset mismatch 😞
- Monotonic switching $\rightarrow$ large V_{cm} variation $\rightarrow$ offset calibration cannot be done at the fixed V_{cm} 😞

Comparator offset foreground calibration

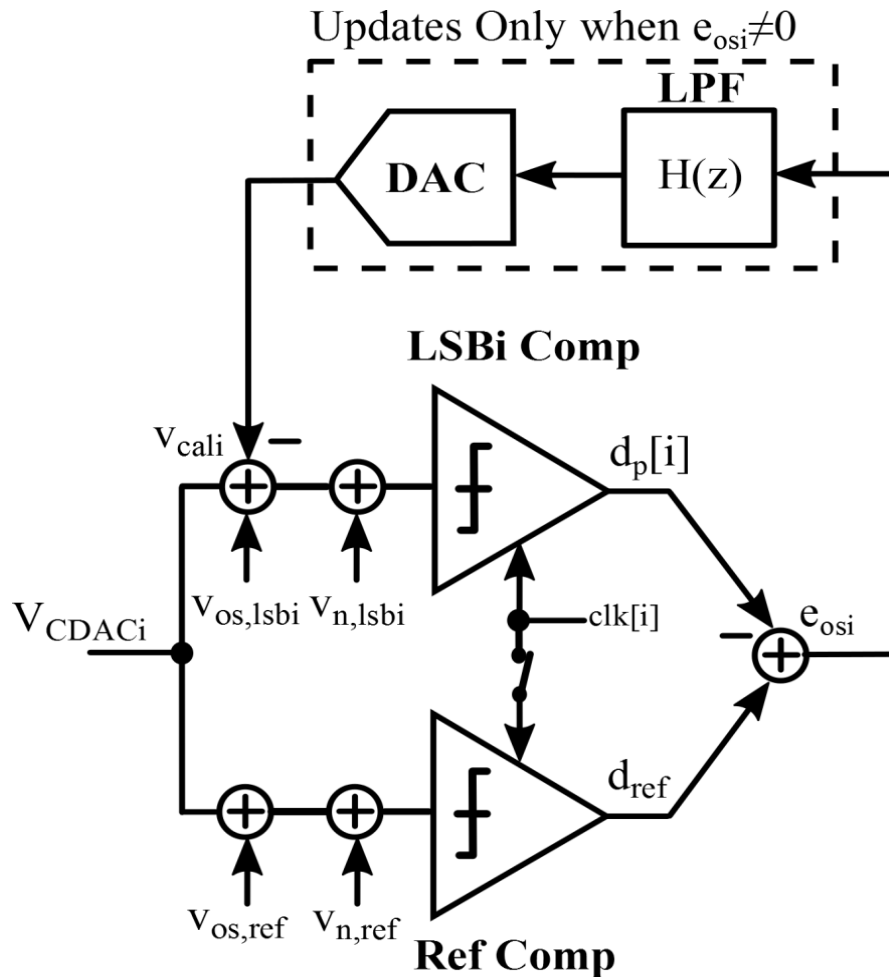


- Bidirectional single-side switching to reduce V_{cm} variation 😊
 - Simple change to the circuit
 - Optimized V_{cm} for speed and noise
- V_{cm} adaptive offset calibration 😊
 - Shorting differential DAC during the calibration mode

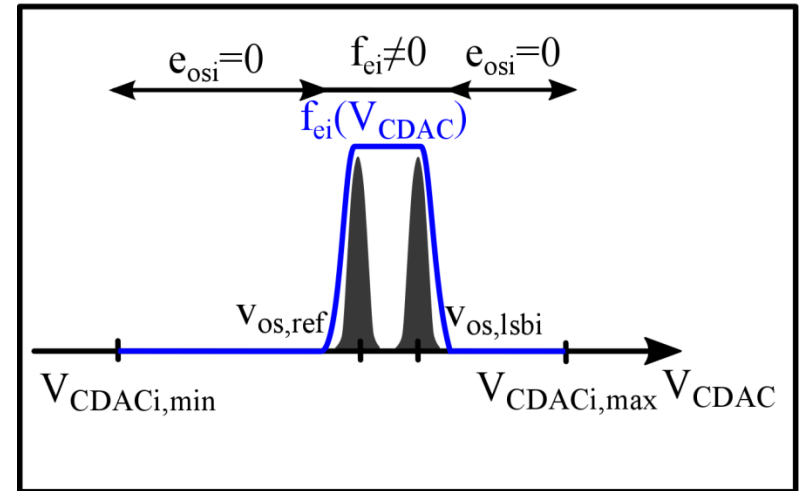
Proposed Architecture



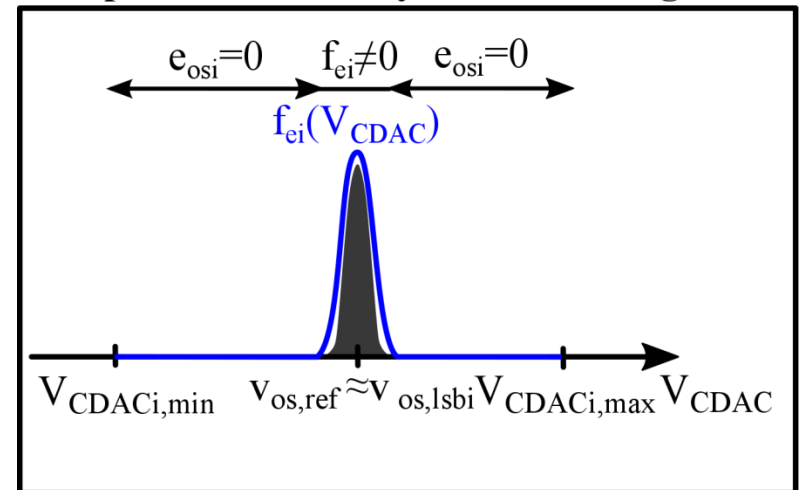
LSB Comparators Background Offset Calibration



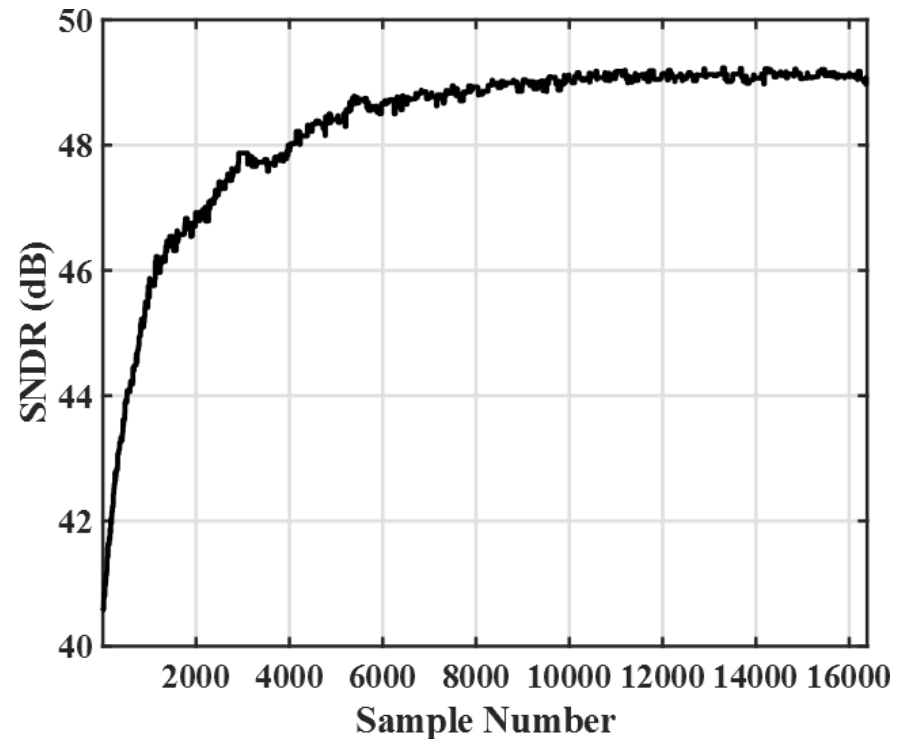
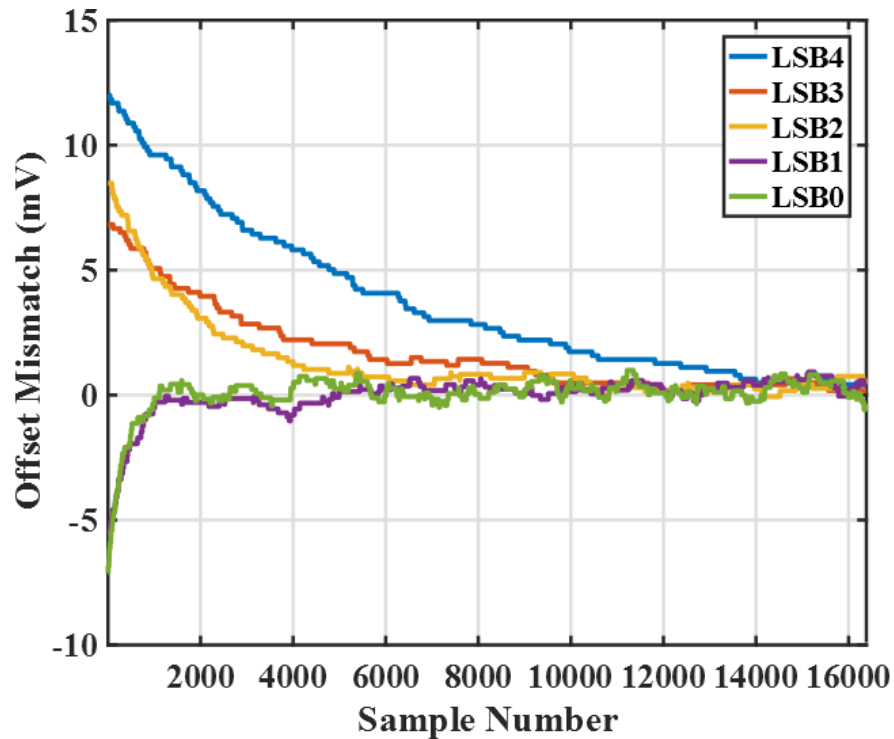
Update Probability at Power-ON



Update Probability after Convergence

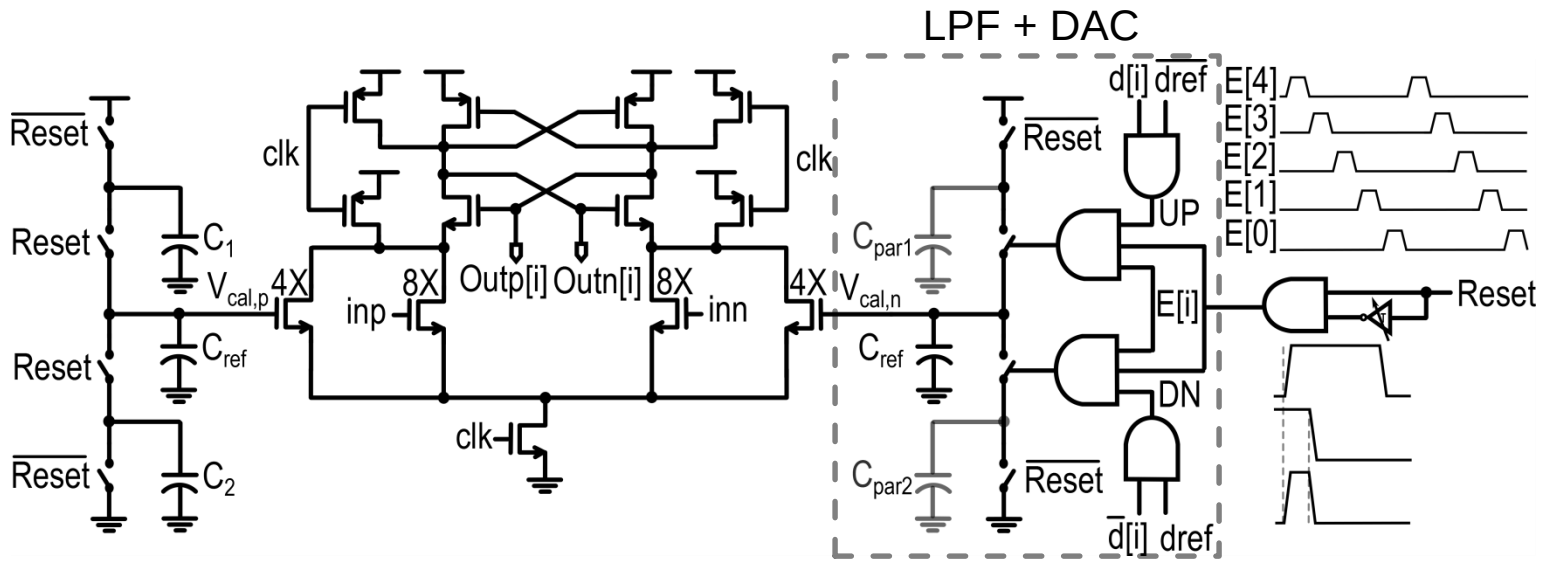


Background Calibration Loop Convergence



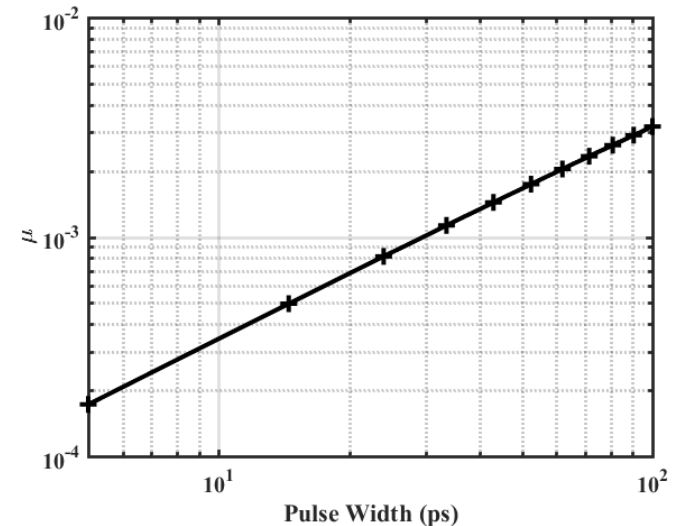
Convergence time $\sim 30\mu\text{s}@350\text{MS/s}$

Circuit Implementation: Compact LPF/DAC

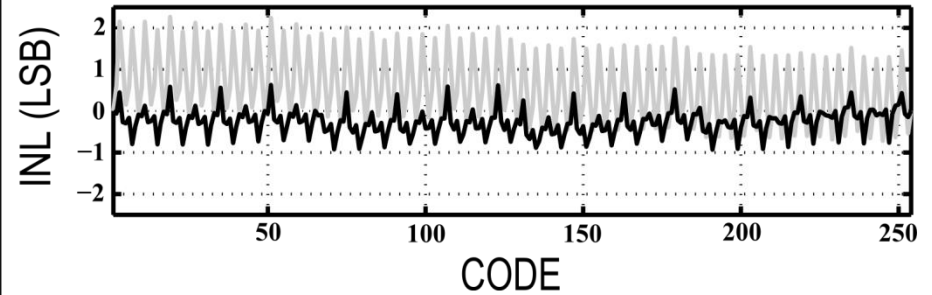
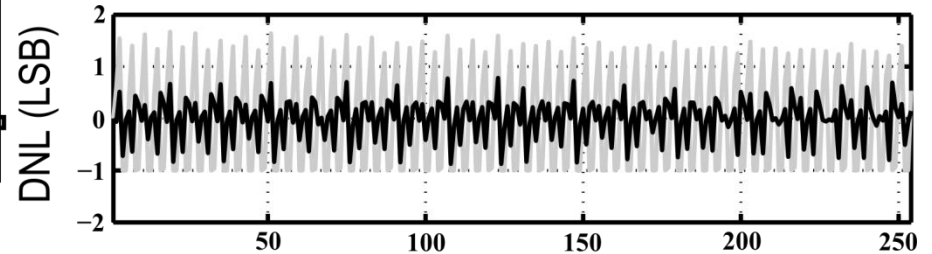
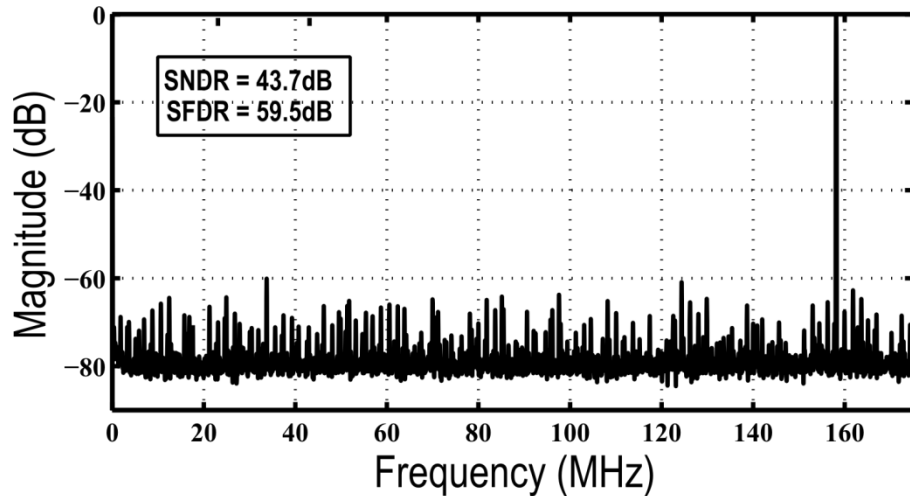
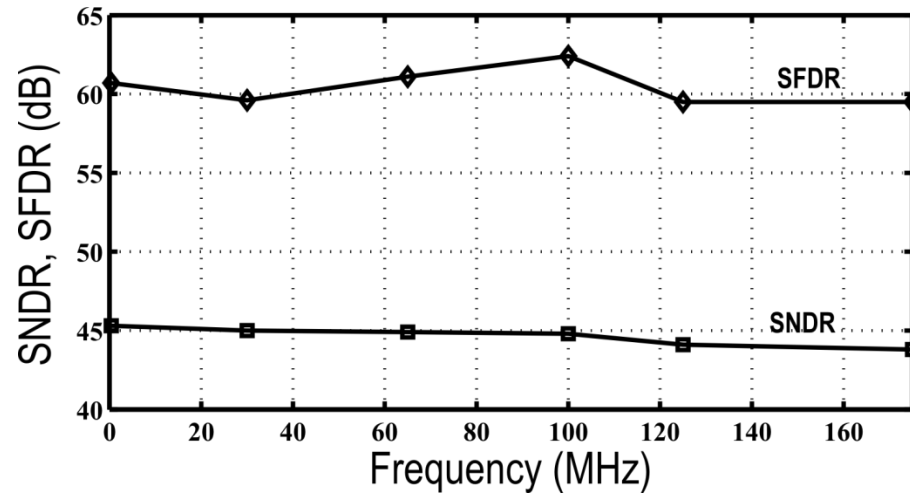


$$V_{cal,n}[n] = \begin{cases} (1 - \mu)V_{cal,n}[n - 1] + \mu \cdot V_{dd}, & d_{ref} = 0 \text{ and } d = 1 \\ (1 - \mu)V_{cal,n}[n - 1] - \mu \cdot V_{dd}, & d_{ref} = 1 \text{ and } d = 0 \\ V_{cal,n}[n - 1], & d_{ref} = d \end{cases}$$

$$\mu = \frac{C_{par}}{C_{cal} + C_{par}} \left(1 - e^{\frac{-t_p}{\tau_{cal}}} \right) \cong \frac{C_{par}}{C_{cal}} \frac{t_p}{\tau_{cal}}$$

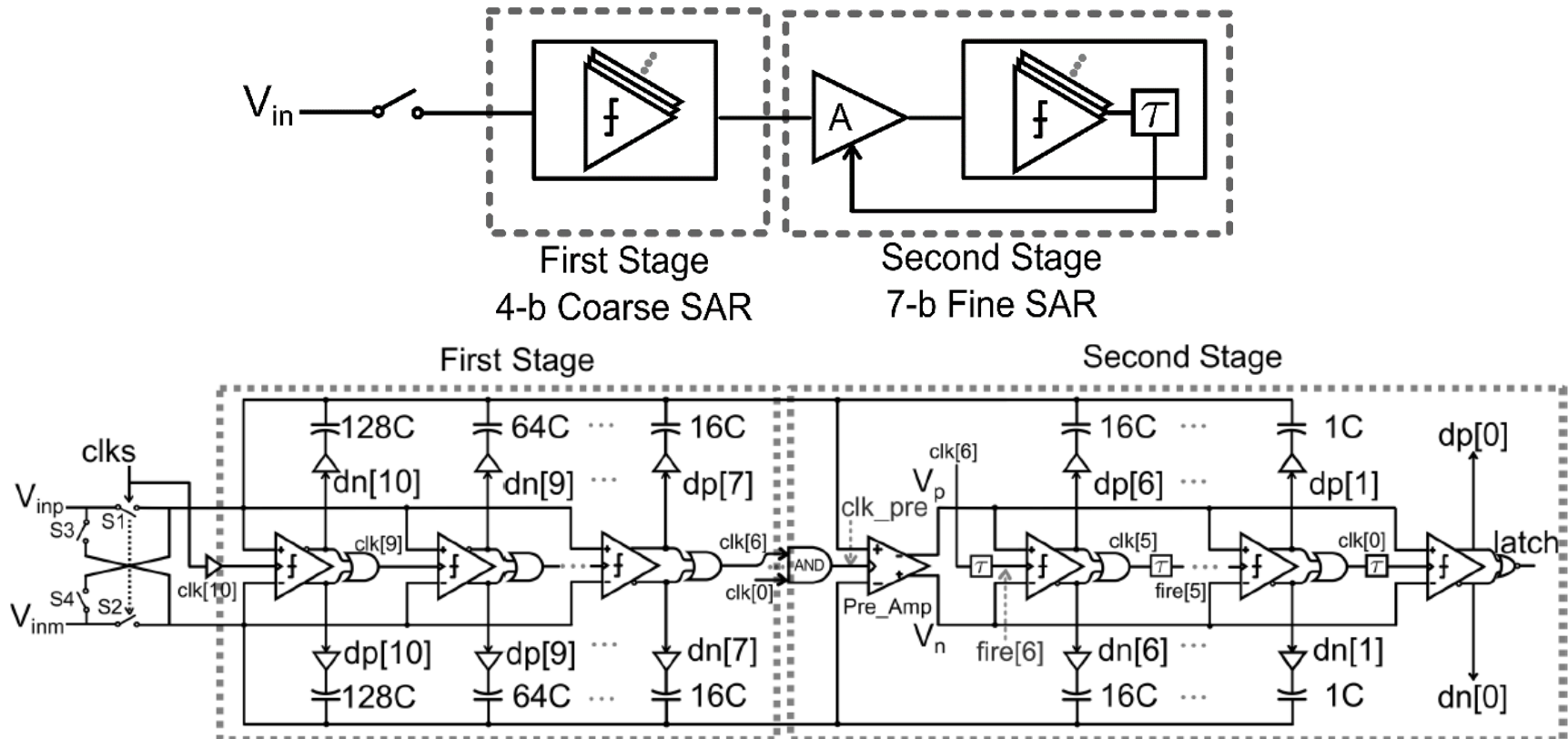


Measurement Results

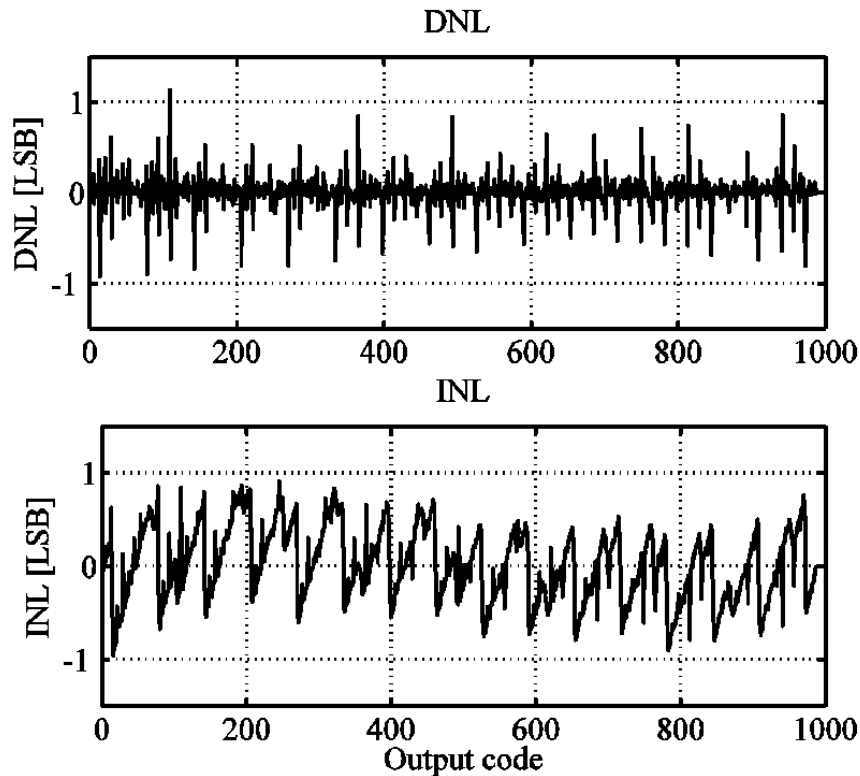


Two-stage sub-ranging architecture

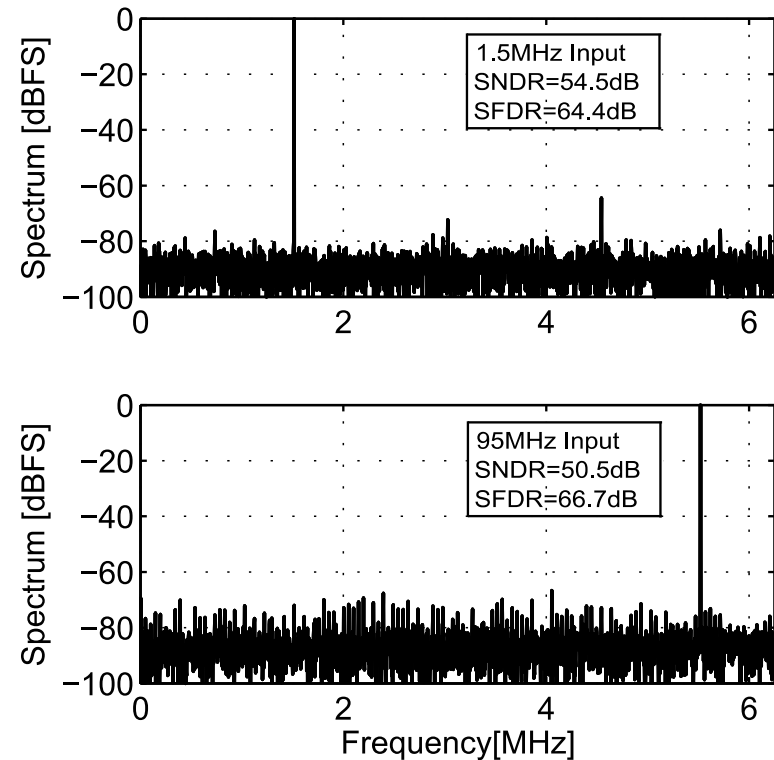
- Two-stage loop-unrolled SAR with shared DAC (no need for interstage gain cal)
 - Pre-amp is shared and attenuates the offset mismatch → Calibration free 😊
 - Fully dynamic 😊 suitable for higher resolution 😊
 - Add pre-amp delay 😊



Measurement Results



DNL: +1.1/-0.9 LSB
INL: +0.9/-1.0 LSB



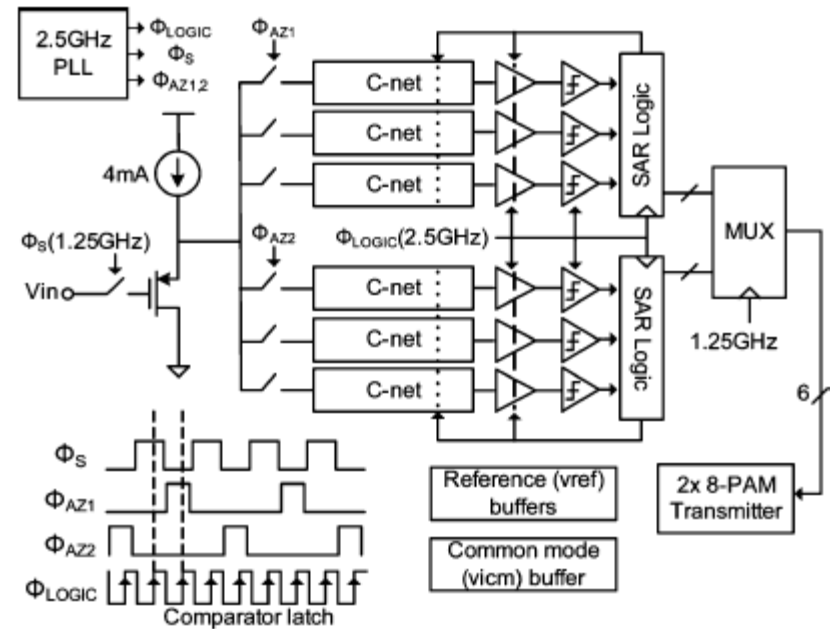
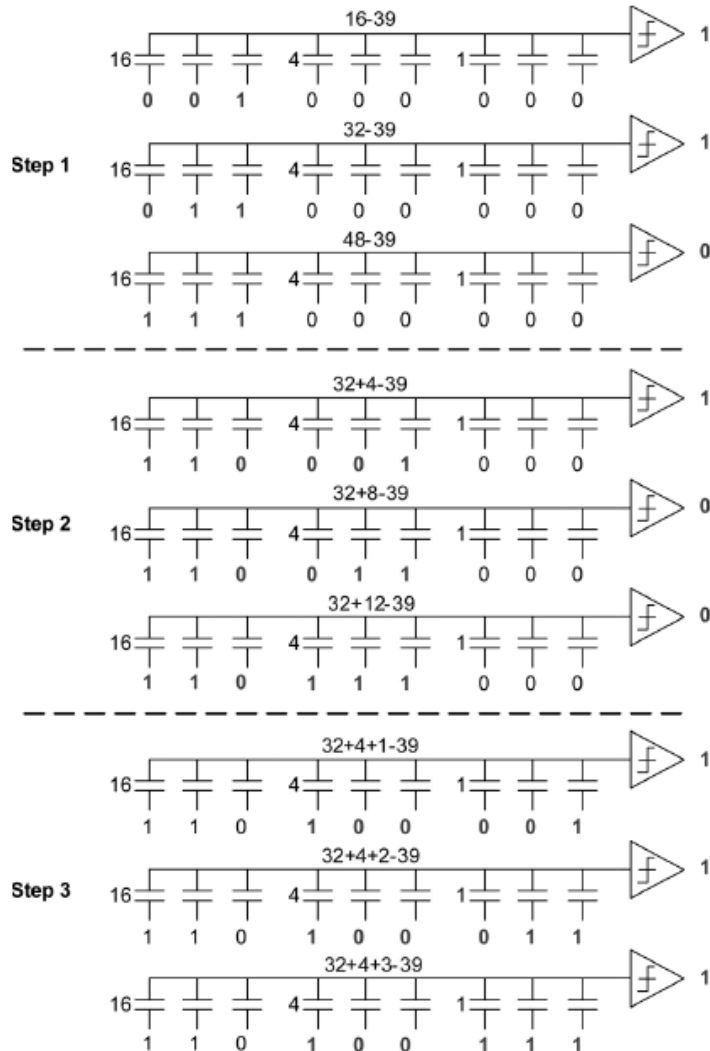
Low freq input: 54.5dB
Nyquist input: 50.5dB

FoM of 8.6fJ/step at 200MS/s

High-speed SAR design

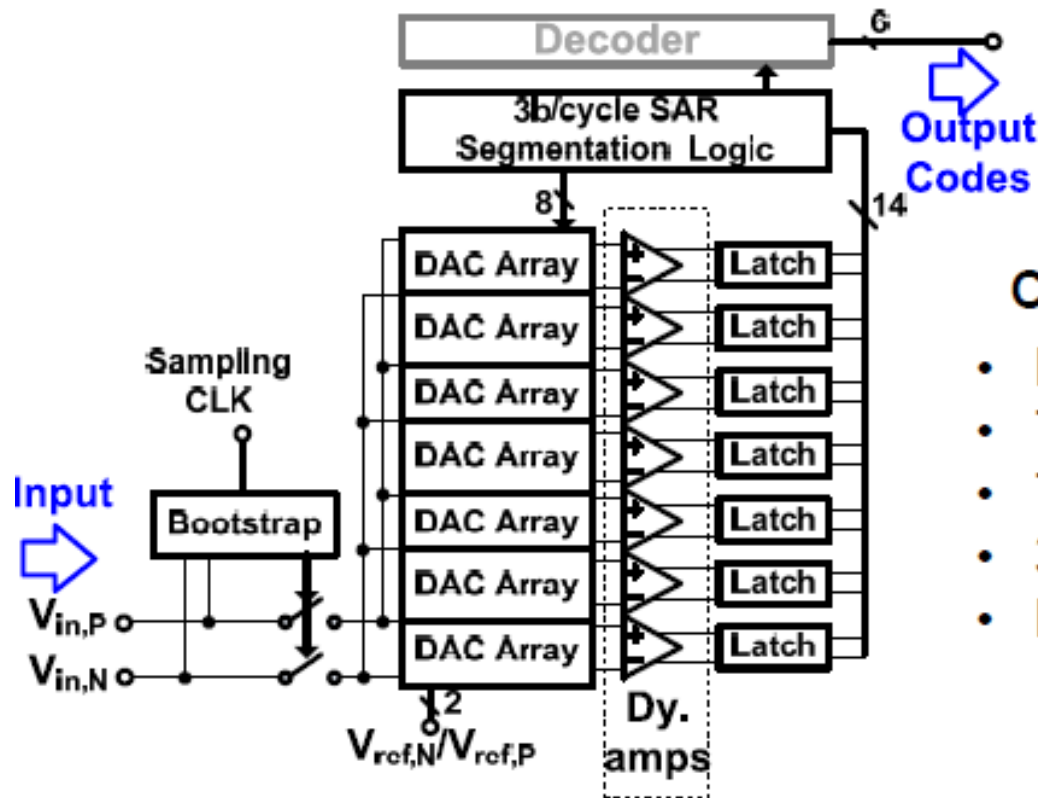
- $T_{loop} = t_{comp} + t_{logic} + \max\{t_{DAC}, t_{reset}\}$
- $T_{SAR} \approx t_{sample} + N \cdot T_{loop}$
- Reducing t_{comp}
 - Asynchronous clocking
- Reducing t_{logic}
 - Direct path from comparator to DAC
- Reducing t_{DAC}
 - Add redundancy
 - Self timing
- Reducing t_{reset} ($t_{reset} > t_{DAC}$ for $b < 6$ -bit)
 - Ping-pong operation
 - Loop unrolled SAR
- Reducing N
 - Multi-bit-per-cycle

Reducing N: 2b/cycle



- Reduced N to N/2 😊
- More DACs and more comparators ☹️
- DAC and comparator offset mismatch ☹️

3b/cycle

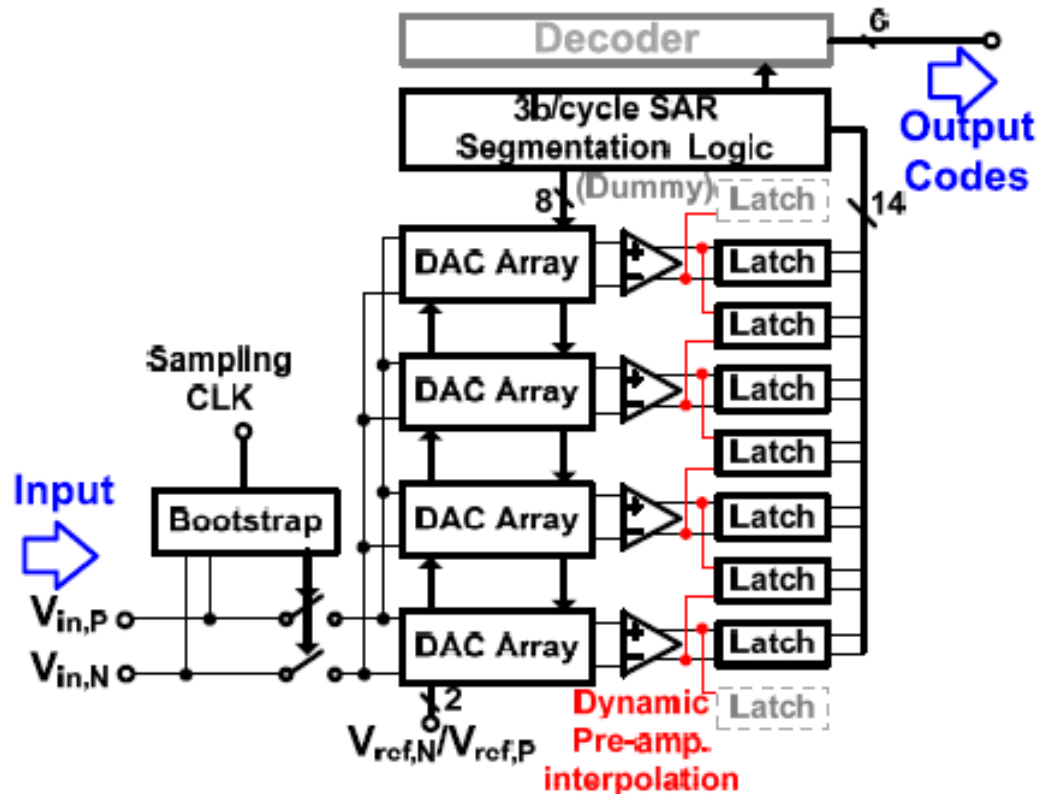


ADC Architecture:
• 3b/cycle SAR

Consist of:

- Bootstrap sampling
- 7 dynamic pre-amps
- 7 DACs & latches
- 3b/cycle logic
- Decoder

3b/cycle



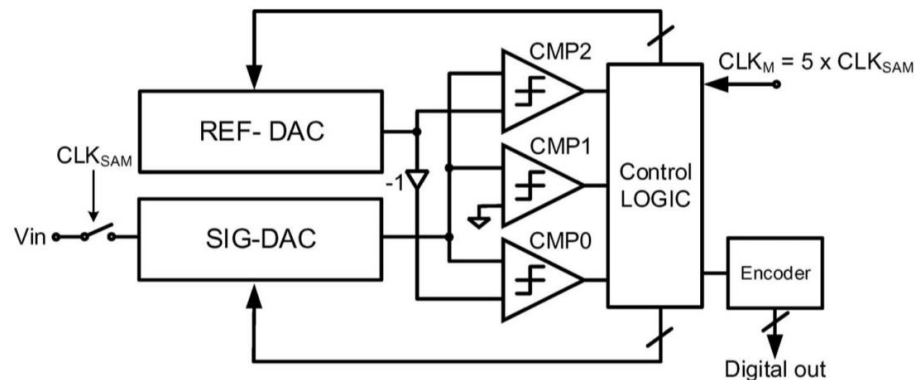
ADC Architecture:

- 3b/cycle SAR

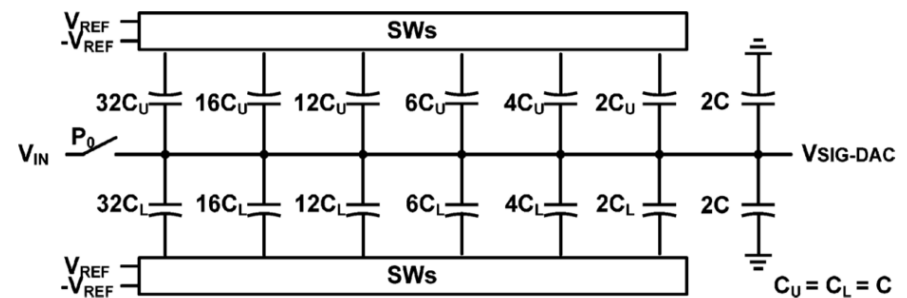
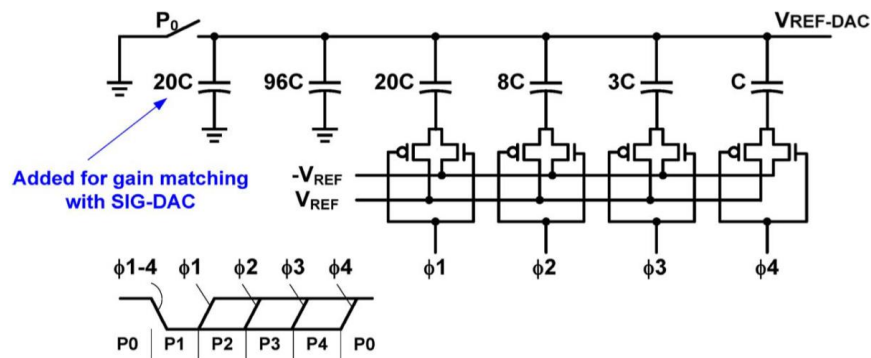
Hardware saving technique:

- Interpolation @ dynamic pre-amps' outputs

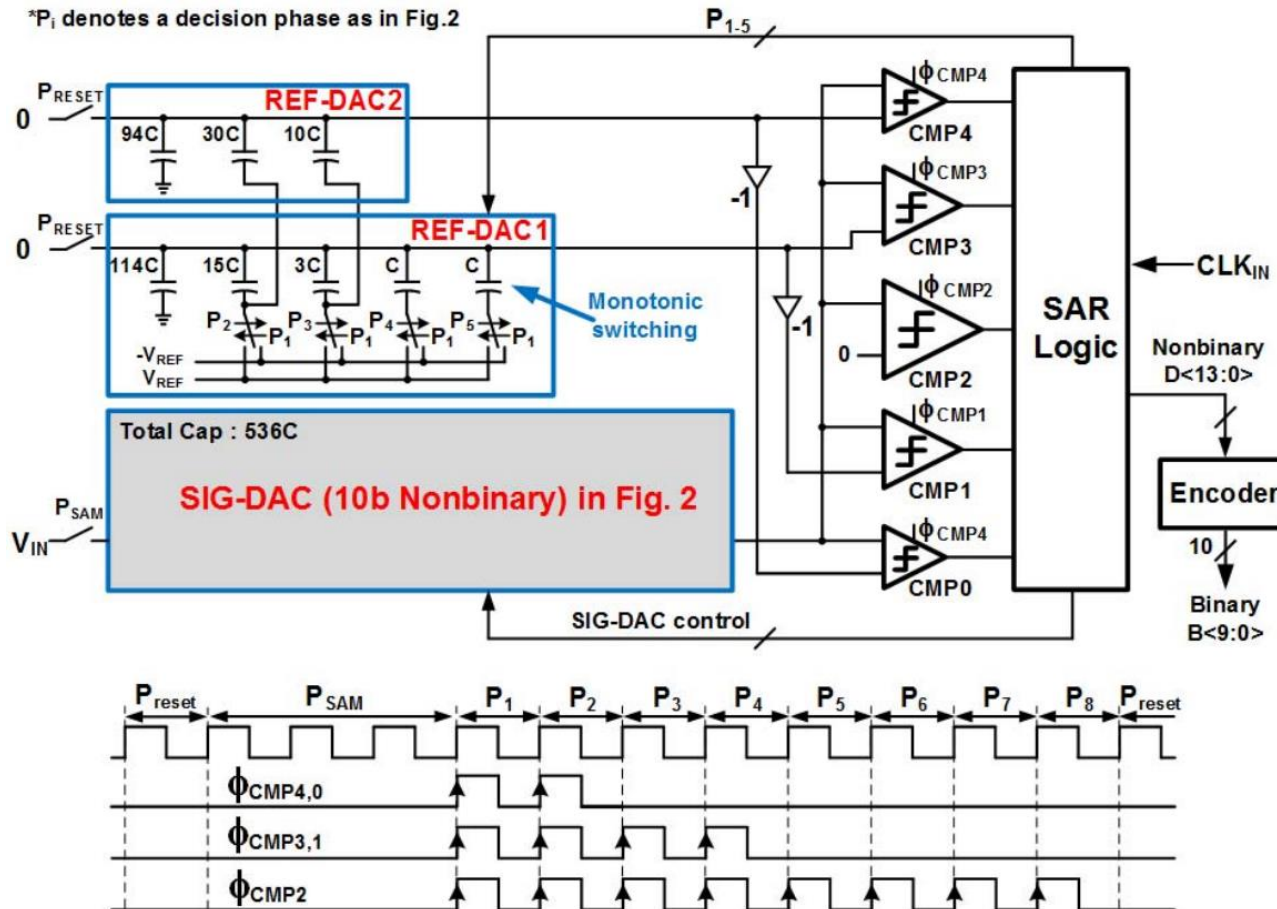
2b/cycle SAR with REF/SIG DAC



- A dedicated DAC for reference generation
- Need only 2 DACs 😊
- Need to consider gain mismatch between DACs ☹️

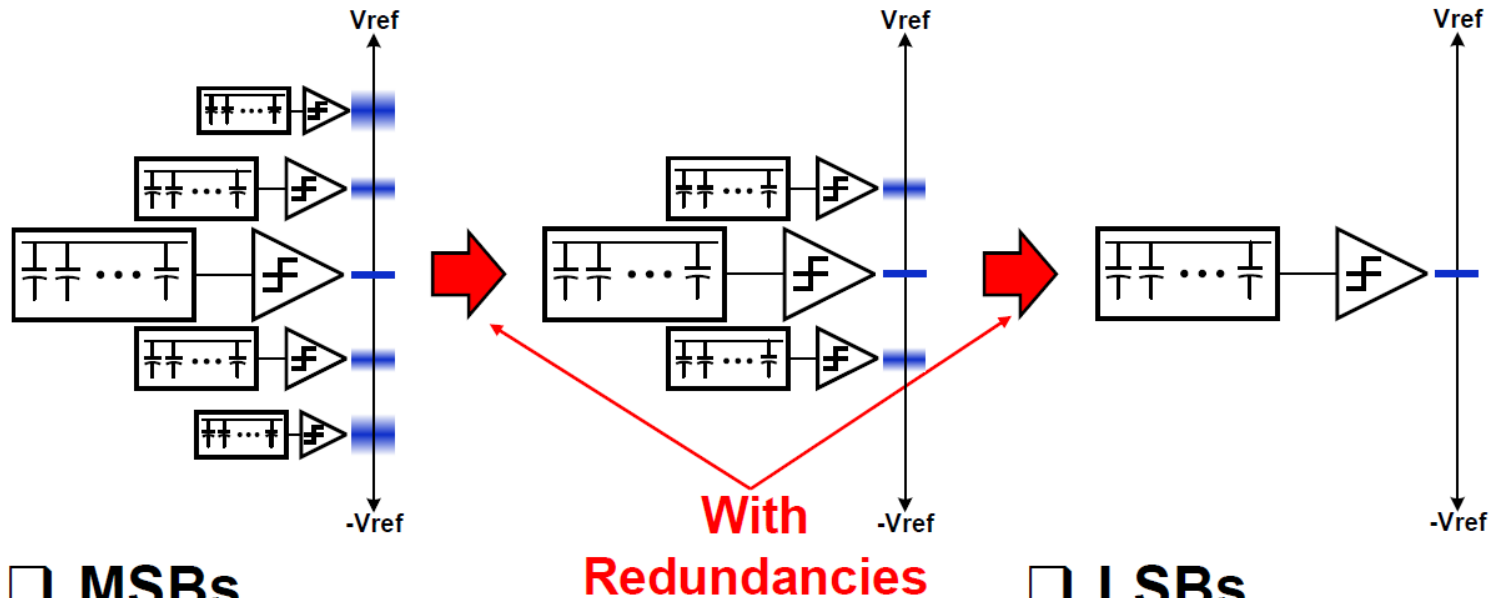


A 2.6b/cycle with a Multistep Hardware-Retirement



- 3 DACs (1 SIG DAC + 2 REF DAC) to achieve 2.6b/cycle

Hardware retirement



❑ MSBs

- Low accuracy
- Multi-bits/cycle
 - High speed

❑ LSBs

- High accuracy
- 1b/cycle
 - Low speed

Use hardware retirement to address gain and comparator offset mismatch ☺

Prior 2b/cycle works

- All requires extra C-DAC or R-DAC to generate reference voltages
 - Requires extra area & power
 - Vulnerable to gain mismatch
 - Requires extra control logics
- Can we achieve 2b/cycle without any extra DAC?
 - 1 differential DAC generate 3 comparison voltage?
 - Prior works use only differential-mode signal
 - Add one more dimension: **common-mode**

Proposed 2b/cycle SAR Algorithm

Proposed Algorithm

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{cmi}$$

$$\Leftrightarrow V_{inp} - V_{inm} \geq -V_{ref}$$

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{inm} + \frac{1}{2}V_{ref}$$

$$\Leftrightarrow V_{inp} - V_{inm} \geq 0$$

$$V_{inm} + \frac{1}{2}V_{ref} \leq V_{cmi}$$

$$\Leftrightarrow V_{inp} - V_{inm} \geq V_{ref}$$

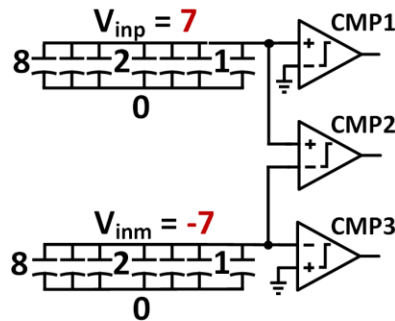
$$* V_{cmi} \equiv (V_{inp} + V_{inm})/2$$

- Left-hand side equations provide 3 comparison equations
- Only need 2 voltages: $V_{inp} + V_{ref}/2$ and $V_{inm} + V_{ref}/2$

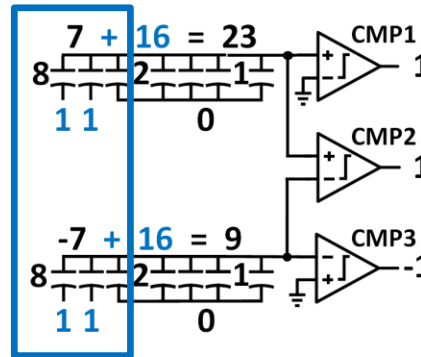
A 5-b example (2 2b/cycle + 1 1b/cycle)

CM updates add $V_{ref}/2$ by raising $C_{total}/2$

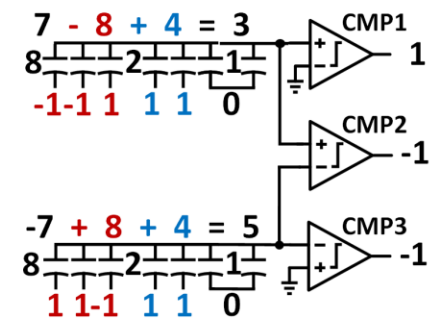
Step 1
Sample



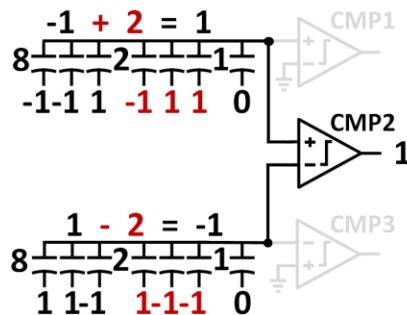
Step 2 (2b/cycle)
Add $\frac{1}{2}V_{ref}$ & comparison



Step 3 (2b/cycle)
Add $\frac{1}{8}V_{ref}$ & comparison



Step 4
Transform to 1b/cycle



CM updates are colored in blue DM updates are colored in red

Implemented Algorithm

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{cmi} \iff V_{inp} - V_{inm} \geq -V_{ref}$$

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{inm} + \frac{1}{2}V_{ref} \iff V_{inp} - V_{inm} \geq 0$$

$$V_{inm} + \frac{1}{2}V_{ref} \leq V_{cmi} \iff V_{inp} - V_{inm} \geq V_{ref}$$

Decoder scheme

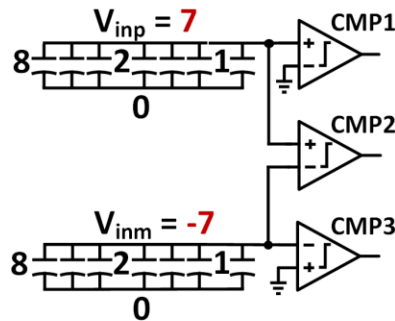
CMP1	CMP2	CMP3	2b
-1	-1/1	-1	-1-1
1	-1	-1	-1 1
1	1	-1	1-1
1	-1/1	1	1 1

ADC output = [1 -1 -1 1 1] $\rightarrow 2^4 - 2^3 - 2^2 + 2 + 1 = 7$

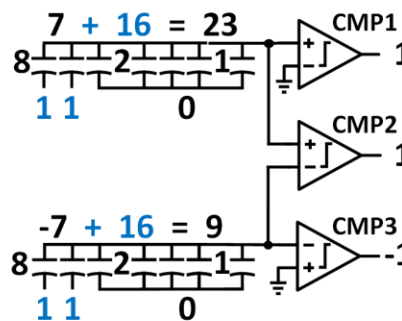
A 5-b example (2 2b/cycle + 1 1b/cycle)

Simultaneously update the DM and CM of C-DAC

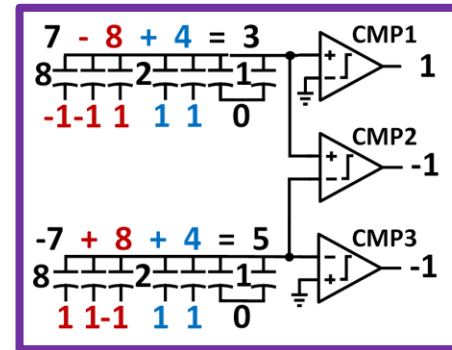
Step 1
Sample



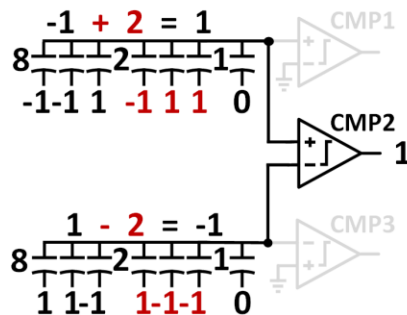
Step 2 (2b/cycle)
Add $\frac{1}{2}V_{ref}$ & comparison



Step 3 (2b/cycle)
Add $\frac{1}{8}V_{ref}$ & comparison



Step 4
Transform to 1b/cycle



CM updates are colored in blue DM updates are colored in red

Implemented Algorithm

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{cmi} \iff V_{inp} - V_{inm} \geq -V_{ref}$$

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{inm} + \frac{1}{2}V_{ref} \iff V_{inp} - V_{inm} \geq 0$$

$$V_{inm} + \frac{1}{2}V_{ref} \leq V_{cmi} \iff V_{inp} - V_{inm} \geq V_{ref}$$

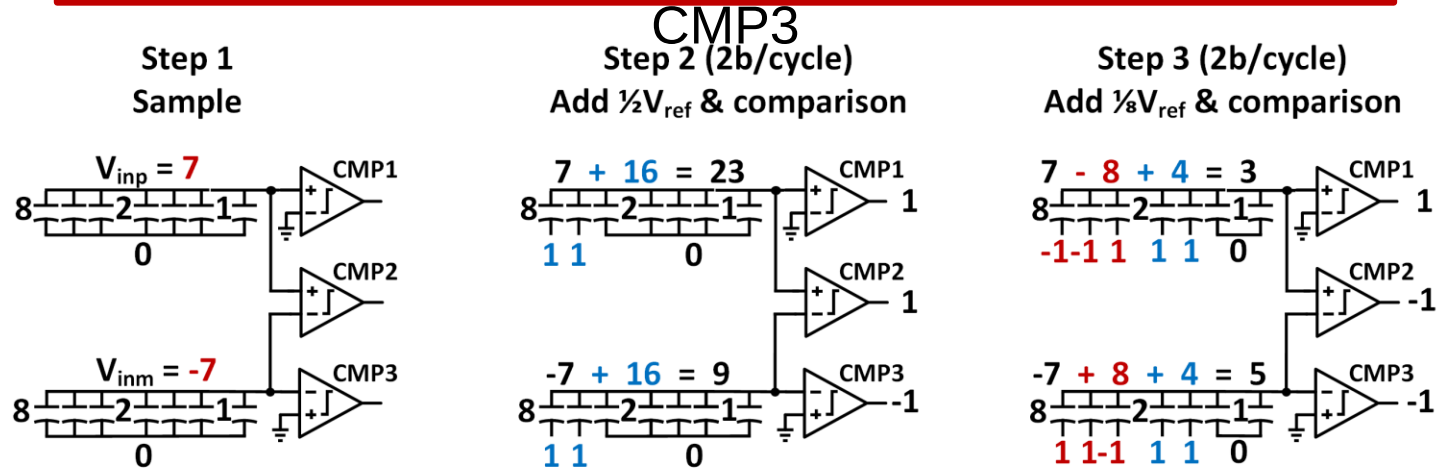
Decoder scheme

CMP1	CMP2	CMP3	2b
-1	-1/1	-1	-1-1
1	-1	-1	-1 1
1	1	-1	1-1
1	-1/1	1	1 1

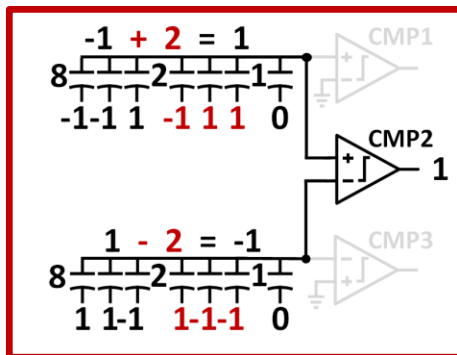
ADC output = [1 -1 -1 1 1] $\rightarrow 2^4 - 2^3 - 2^2 + 2 + 1 = 7$

Hardware retirement to enhance resolution

Simple conversion to 1b/cycle by disabling CMP1 and



Step 4
Transform to 1b/cycle



CM updates are colored in blue DM updates are colored in red

Implemented Algorithm

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{cmi} \iff V_{inp} - V_{inm} \geq -V_{ref}$$

$$V_{inp} + \frac{1}{2}V_{ref} \geq V_{inm} + \frac{1}{2}V_{ref} \iff V_{inp} - V_{inm} \geq 0$$

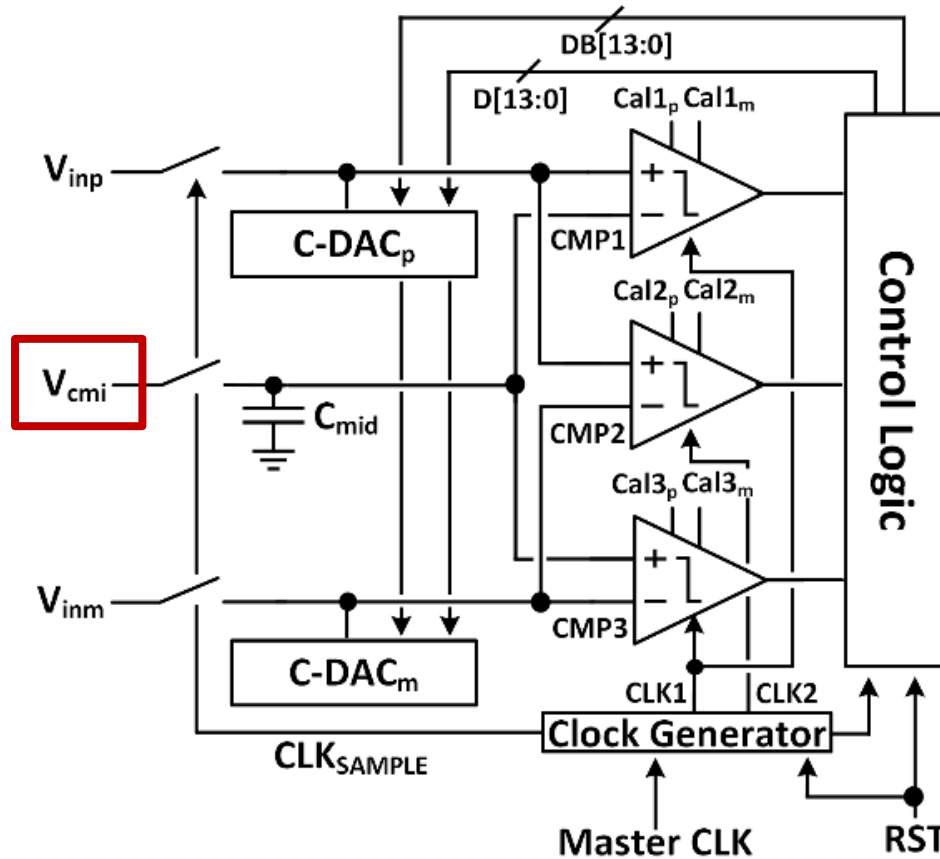
$$V_{inm} + \frac{1}{2}V_{ref} \leq V_{cmi} \iff V_{inp} - V_{inm} \geq V_{ref}$$

Decoder scheme

CMP1	CMP2	CMP3	2b
-1	-1/1	-1	-1-1
1	-1	-1	-1 1
1	1	-1	1-1
1	-1/1	1	1 1

ADC output = [1 -1 -1 1 1] $\rightarrow 2^4 - 2^3 - 2^2 + 2 + 1 = 7$

Circuit implementation



- V_{cmi} is assumed to be available. It can be generated internally through a voltage divider
- V_{cmi} can also be generated inside the ADC and sent out to the ADC driver as the output common-mode reference

Re-cap on high-speed SAR

- A few high-speed SAR ADC design techniques are presented
 - Need to tackle every aspect in the delay
- SARs are very fast now
 - Single channel can be $> 1\text{GS/s}$
 - In fact, the fastest ADCs ($\sim 100\text{GS/s}$) are made by time-interleaved SAR
- Made possible by not only technology scaling, but also circuit innovations